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# Probabilistic Modeling and Statistical Inference

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Decision making is a critical task that spans not just science and industry but arguably the entire human experience. At the same time, decision making in complex applications is notoriously subtle and counterintuitive.

How do we make a principled selection between a set of actions,  $A$ ? This decision becomes straightforward if each possible action is assigned a *utility*,

$$U : A \rightarrow \mathbb{R}^+ \\ a \mapsto U(a),$$

that quantifies the overall benefit of that action minus any costs. Given such a utility assignment we could then simply select the action with the highest utility,

$$a^* = \operatorname{argmax}_{a \in A} U(a).$$

In practice, however, even the most rigorous utility assignments will depend on circumstances we do not know with perfect certainty. Here we must rely on *inferences* to inform those circumstances and motivate effective decision making.

In this case study we'll review the foundations of statistical models and statistical inference that advise principled decision making. We'll place a particular emphasis on Bayesian inference, which utilizes probability theory and statistical modeling to encapsulate information from observed data and our own domain expertise.

I take an observational perspective here that presumes the existence of a true data generating process that generates the observations from which we learn. This perspective is not universal and clashes with some philosophical perspectives on statistics that are not uncommon in popular introductory texts. Fortunately, in most cases the differences between these perspectives don't have any consequences for how the resulting methodology is implemented in practice. That said, it helps to keep in mind the differing perspectives when comparing this material to other literature. Prevalent frequentist texts include [1], [2] and [3]. There are a wide array of introductory Bayesian texts from objective Bayesian perspectives [4], subjective Bayesian perspectives [5], and more applied perspectives [6].

This material will presume a familiarity of probability theory at the level introduced in my probability theory ([https://betanalpha.github.io/assets/case\\_studies/probability\\_theory.html](https://betanalpha.github.io/assets/case_studies/probability_theory.html)) and conditional probability theory ([https://betanalpha.github.io/assets/case\\_studies/conditional\\_probability\\_theory.html](https://betanalpha.github.io/assets/case_studies/conditional_probability_theory.html)) case studies.

## 1 Probabilistic Modeling

Models are approximations of the complex dynamics that drive the observable phenomena in the world around us. They provide the setting in which we can formalize learning and decision making and hence are a foundational aspect to any rigorous analysis. In this section we review latent observational processes, the data generating processes they induce, and the statistical models that offer a spectrum of possible data generating behaviors.

### 1.1 The Observational Process

A typical abstraction in applied analysis is to assume that we are making decisions about how to interact with some latent *phenomenon*. Our inferential target is then the phenomenon itself.

No material phenomenon, however, exists in isolation. Any phenomenon relevant to statistical inquiry instead manifests within some latent system that includes, at the very least, the mechanisms with which we can interact with it. In other words we can observe the phenomenon of interest only through its surrounding *environment*.

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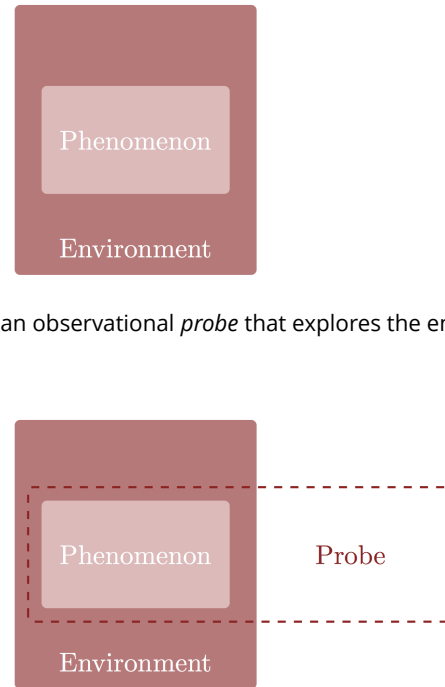
In order to make the desired inferences we need an observational *probe* that explores the environment and, hopefully, the phenomenon of interest.

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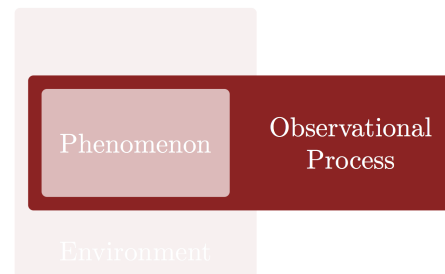
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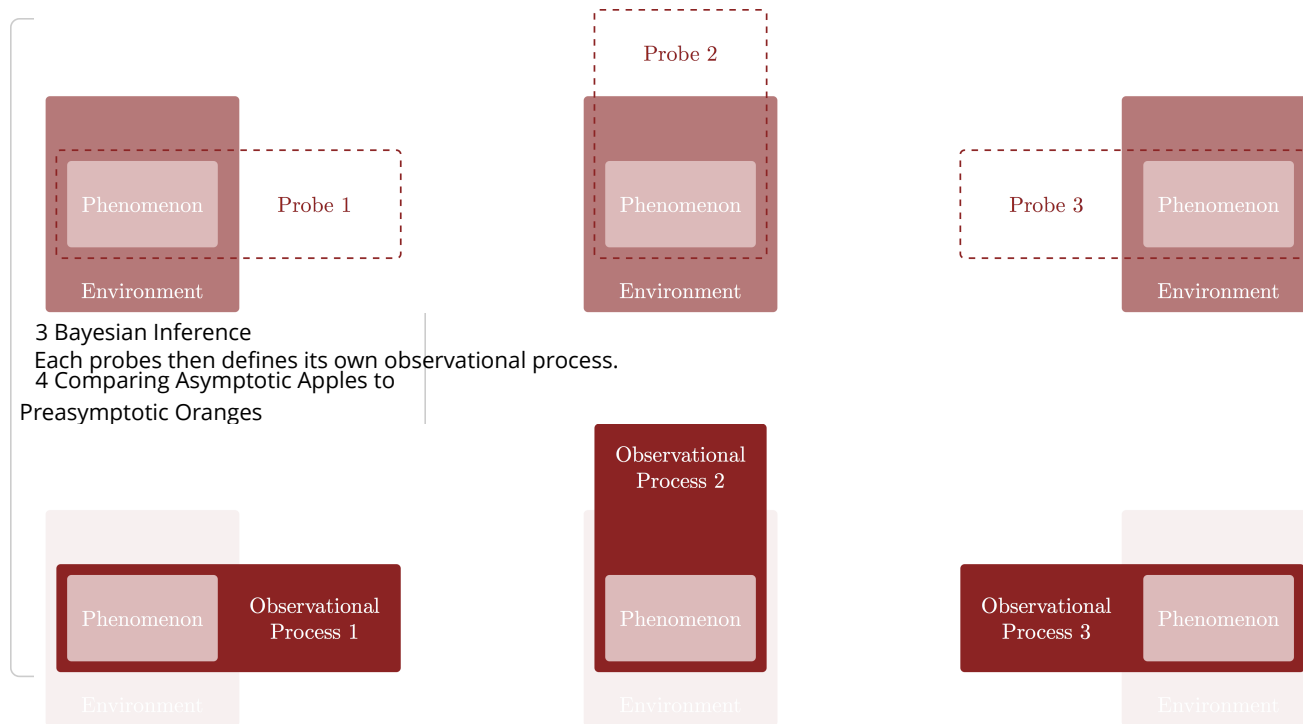
The interaction of the probe with the environment and the phenomenon defines an *observational process* which examines the latent system and induces explicit observations that are sensitive to the phenomenon of interest.



An observational process encompasses any methodology that measures, collects, curates, or manipulates observations sensitive to the latent phenomenon. This includes, for example, not only dedicated experiments explicitly designed to interrogate the phenomenon of interest but also retrospective analyses of incidental experiments or previously conducted experiments designed for other purposes.

In encapsulating the phenomenon of interest our probe intersects only part of the surrounding environment. Consequently the observational process will be sensitive to only those elements of the environment relevant to the probe. Well-designed experiments offer the promise of better isolating the target phenomenon, yielding a much cleaner observational process and more informative observations.

In general there is no unique procedure for interrogating an environment and the phenomena within it. In other words, there is never a singular experimental probe. Instead each of the many possible procedures results in its own probe.



3 Bayesian Inference  
 Each probes then defines its own observational process.  
 4 Comparing Asymptotic Apples to  
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These distinct observational processes can become the basis of independent analyses, or even components in a single joint analyses that aggregates data from multiple sources.

Consequently observational processes are bespoke to not only a given phenomenon but also the specific details of the probe used to investigate it, especially the environment with which the probe intersects. A critical aspect of statistical modeling is capturing enough of these details to isolate inferences about the latent phenomenon that can then be fairly compared across different experiments. This is especially true when the specific behavior of the environment or the probe is transient or otherwise not exactly repeatable. Without a rich understanding of the observational process phenomenological inferences can poorly generalize outside of the exact context of the given experiment, promoting more confusion than comprehension.

## 1.2 The True Data Generating Process

By definition an observational processes is only a conceptual abstraction. In order for it to take form as an explicit mathematical object we need to define exactly how observations are *realized* from an observational process. This process, however, is subtle and results not in constant outcomes but rather a *variation* of possible outcomes within an *observation space*,  $Y$ .

For example the realized observations might exhibit variation due to physical, or *ontological*, stochasticity inherent to the latent phenomenon, its environment, or the probe used to interrogate them. This variation is also known as *aleatoric randomness* of the observational process. Ontological variation, for example, is commonly posited when modeling atomic decays that produce radiation at varying intervals and in varying directions, or when sampling individuals from a population.

At the same time the realized observations might also exhibit *epistemological* variation due to limitations in how accurately we can observe the latent system. Lengths measured with a ruler, for example, are limited by the precision of the markings and the resolution of the human eye; they will not in general be consistent from person to person or even a single person

from time to time. Similarly, increasingly common digital data collection is limited by the discretization of analog inputs and the numerical precision of intermediate computations.

## 1 Probabilistic Modeling

Epistemological variation is particularly common in the study of dynamical systems. Chaos theory, for example, arises when infinitesimal uncertainty in initial conditions is evolved through nonlinear, but deterministic, dynamics. This evolution results in probabilistic behavior of the system despite its fundamental determinism. Likewise thermodynamics is a theory of probabilistic behavior in many-particle systems that emerges from deterministic microscopic dynamics when only certain macroscopic quantities can be observed.

### 1.4 Model-Based Inferences

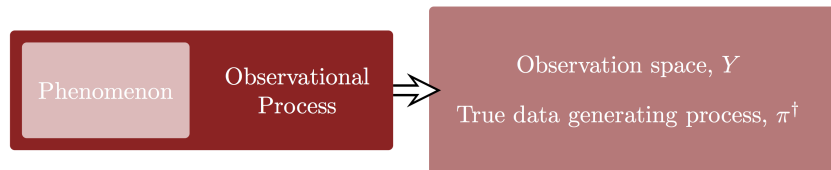
Ultimately, however, the distinction between these two sources of variation is ill-defined. For example, one can argue that seemingly ontological variation is a result of the imprecise observation of complex but deterministic dynamics, as in chaos theory. Likewise one can argue that epistemological variation defines an ensemble of possible but unobserved, or *counterfactual*, outcomes which manifests as ontological variation if the experiment were hypothetically repeated.

### 4 Comparing Asymptotic Apples to

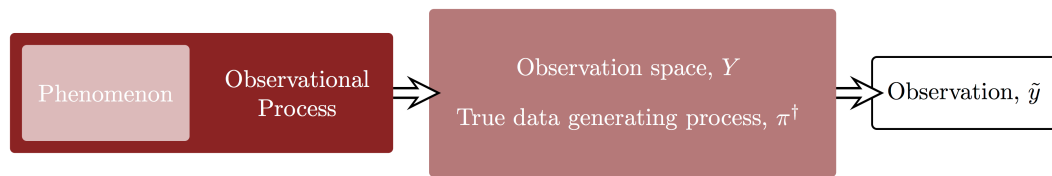
Fortunately, the exact philosophical nature of this variation isn't relevant in practice because these different perspectives all manifest in the *same* mathematical assumptions for the realization of the observational process. Regardless of its

interpretation we presume that this variation is sufficiently regular that, while an observational process cannot be quantified by any single observation, it can be quantified with a *distribution* of observations. More formally we model the observational process with a probability distribution over the observation space, with realizations of the observation process modeled as exact samples from that distribution.

Probability distributions over the observation space are in general denoted *data generating processes*, and the probability distribution that exactly captures the observational process in a given application is denoted the *true data generating process*,  $\pi^\dagger$ .



To distinguish possible outcomes of an observational process from explicitly realized observations I will decorate the latter with a tilde. In other words  $y$  denotes an arbitrary point in the observation space while  $\tilde{y}$  denotes a particular point.



#### 4 Comparing Asymptotic Apples to

#### Preasymptotic Oranges

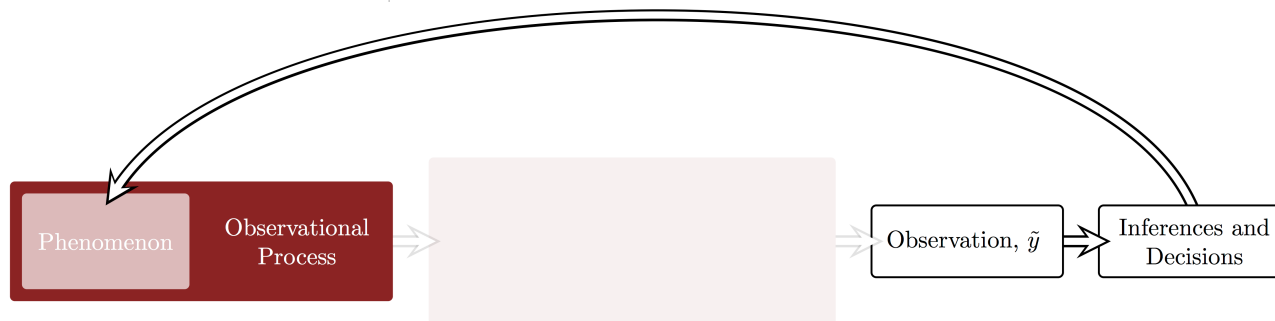
If we have any uncertainty about the system that we're analyzing then the true data generating process will be unknown.

Inference is tasked with quantifying information about the true data generating process using observed data as its only

expression. In particular, successful inferences advice decisions about how to interact with the latent phenomenon

encompassed within the true data generating process.

References



Ignorance of the true data generating process also complicates the *calibration* of any decision making process informed by our inferences. Consider formalizing a decision making process as a map from the observation space to the action space,

$$\begin{aligned} a : Y &\rightarrow A \\ y &\mapsto a(y). \end{aligned}$$

In other words, a decision making process consumes an observation and produces an explicit decision. Given knowledge of the latent system we could hypothetically construct a utility function for each possible action,  $U : A \rightarrow \mathbb{R}^+$ , which would then quantify the benefit of the decision making process in light of a particular observation as

$$\begin{aligned} U \circ a : Y &\rightarrow \mathbb{R}^+ \\ y &\mapsto U(a(y)). \end{aligned}$$

If we knew the true data generating process then we could calibrate the decision making process by evaluating the outcome utilities corresponding to possible observations. More formally, we could push the true data generating process through this composite utility function to give a probability distribution over outcome utilities,  $\pi(U)$ , and then use the shape of this utility distribution to determine the effectiveness of the decision making process.

## 1 Probabilistic Model

### 1.1 The Observational Model

### 1.2 The True Data Generating Process

### 1.3 The Observational Model

### 1.4 Model-Based Calibration

## 2 Frequentist Inference

## 3 Bayesian Inference

## 4 Comparing Approaches

## Preasymptotic Optimality

## 5 Conclusion

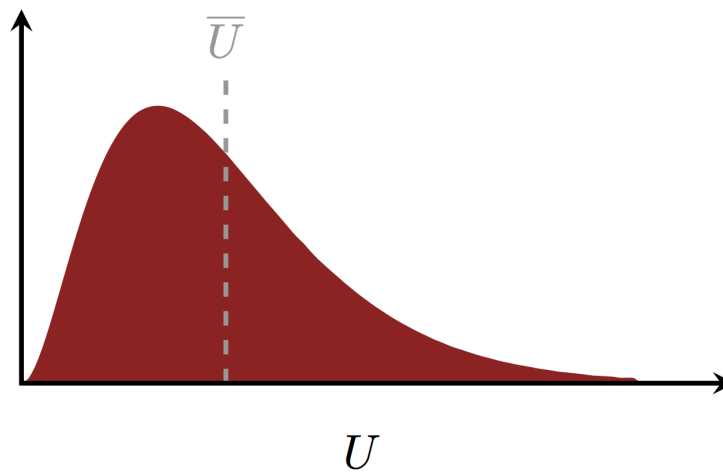
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For example, we could calibrate the decision making process based on its expected utility,

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$$\overline{U} = \int dy \pi^\dagger(y) U(a(y)),$$

Unfortunately this calibration is purely hypothetical because of our ignorance of the true data generating process. In any realistic application we will be able to calibrate a given decision making process only in the context of a statistical model.

## 1.3 The Observational Model

At this point we have presumed the existence of a true data generating process, but we are ignorant of its precise form in any given application. In order to find the appropriate data generating process we would ideally search through the space of *all* data generating processes, that is the collection,  $\Omega$ , of all probability distributions over the observation space.

Unfortunately this *big world* of data generating processes is just too large and complex to be able to quantify in any practical application. Indeed the depths of this space are where pure mathematicians lurk, and that alone should be reason enough to stay away...

In any case, we have to limit our search to something more tractable in practical applications. That is we have to consider only a *small world*,  $\Omega' \subset \Omega$ , of data generating processes that is more manageable. This small world defines an *observational model* consisting of individual data generating process that each define a possible mathematical narrative of how the data could be generated.

## 1 Probabilistic Models

### 1.1 The Observational Model

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### 1.4 Model-Based Inference

## 2 Frequentist Inference

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## 4 Comparing Asymptotic and Preasymptotic Inference

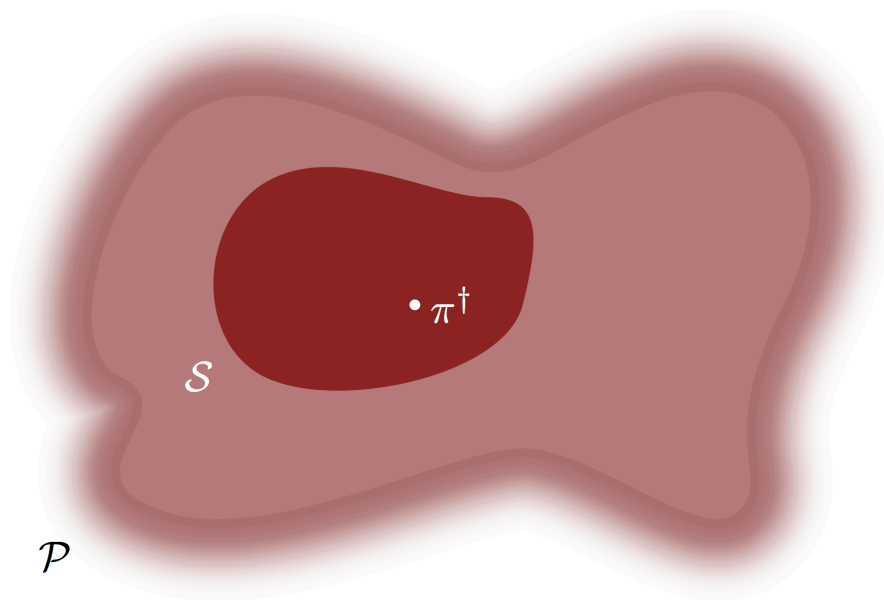
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As the term “small world” often brings with it connotations of theme park rides, I’m going to give this collection of data generating processes the alternative, slightly more sophisticated name of *model configuration space*. Each individual data generating process within the small world is similarly denoted a *model configuration*. In other words “small world”, “observational model”, and “model configuration space” can all be used interchangeably to denote the subset of data generating processes considered in a given analysis. I will use “observational model” when I want to emphasize that a given subset has been assumed for a given application.

In practice it helps to assign distinct, numerical labels to each data generating processes within an observational model. A *parameterization* of an observational model is the one-to-one mapping between the model configuration space and a numerical space  $\Theta$ , such as the real numbers. In other words, a parameterization is the the assignment of a unique value, or *parameter*  $\theta \in \Theta$ , to each model configuration,  $s \in \square$ . Keep in mind that the parameterizations themselves are not unique; in general there will be many equivalent ways of labeling the model configurations.

Given a parameterization we can then specify the observational model with family of probability density functions over the observational space,  $\pi_{\square}(y; s)$ . In practice we index these densities a particular parameterization,  $\pi_{\square}(y; \theta)$ .

For example, let’s say that our observational space is the one-dimensional real numbers,  $Y = \mathbb{R}$ , and our observational model consists of all distributions that admit Gaussian probability density functions over the observational space. We can parameterize this model configuration space with the two-dimensional real numbers,  $\Theta = \mathbb{R}^2$  consisting of *location parameter*,  $\mu$ , that can take all values and a *scale parameter*,  $\sigma$ , that can take only positive values. In this case our model configuration space is specified by

$$\pi_{\square}(y; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2\right).$$

We can equivalently, however, parameterize the model configuration space with a location parameter,  $\mu$ , and an unconstrained scale parameter,  $\lambda = \log(\sigma)$ . In this case our model configuration space is specified by

$$\pi_{\square}(y; \mu, \lambda) = \frac{1}{\sqrt{2\pi}} e^{-\lambda} \exp\left(-\frac{1}{2}(y-\mu)^2 e^{-2\lambda}\right).$$

Under  $N$  repeated observations our observational space would become  $Y = \mathbb{R}^N$  with the corresponding probability

density functions.

## 1 Probabilistic Modeling

### 1.1 The Observational Process

### 1.2 The True Data Generating Process

$$\pi(y_1, \dots, y_N; \mu, \sigma) = \prod_{n=1}^N \pi(y_n; \mu, \sigma).$$

Once we limit our consideration to a given observational model, our inferential goal becomes the quantification of those

model configurations that are somehow similar to the true data generating process, and hence useful for informing our

### 1.4 Model-Based Inferences

ultimate decisions. If the observational model contains the true data generating process then basing our inferences on the

observational model incurs no loss; an exhaustive search through the observational model will always be able to find the

true data generating process. We are not always, however, so fortunate. What happens when the true data generating

### 3 Bayesian Inference

process lies outside the scope of the observational model?

### 4 Comparing Asymptotic Apples to

Preasymptotic Oranges

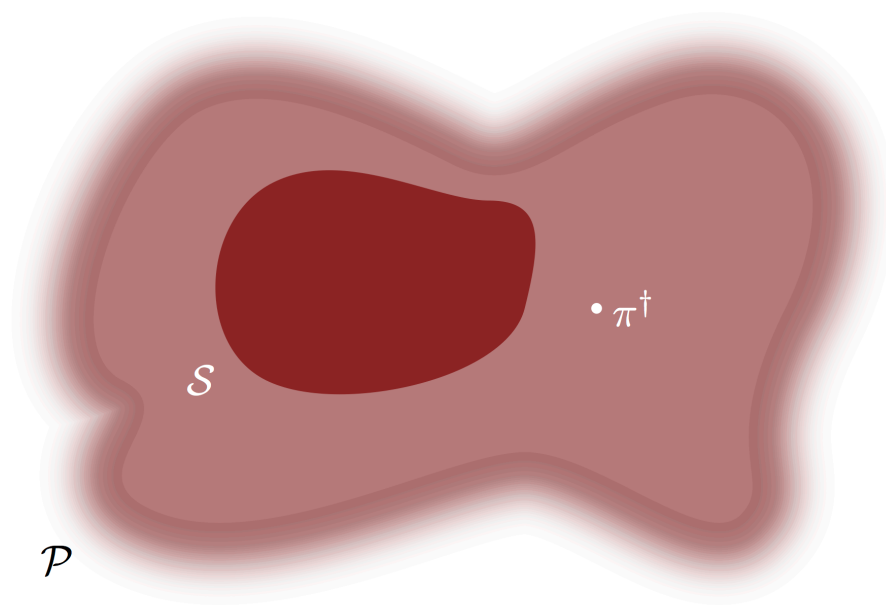
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Indeed in practice this circumstance is the rule rather than the exception. In applications observational models are limited by the constraints of mathematical and computational feasibility, and will not completely capture the, frankly ridiculous, complexity of the natural world.

The inferences generated from an observational model that doesn't contain the true data generating process may still be meaningful so long as the data generating processes it does contain capture the structure of the true data generating process *relevant* to our ultimate inferential goals. For example, an ecological model that neglects quantum mechanics and general relativity is too technically meager to capture the *full reality* of any observation of animal migration. Most ecological research, however, is sufficiently useful without having to consider what might happen if birds started migrating at the speed of light or quantum tunneling through geographical impediments. At the same time, research considering bird navigation guided by following physiological sensitivity to magnetic fields might need some consideration of quantum mechanics.

Ultimately inference within an observational model will at best *approximate* the true data generating process, a realization that resonates with the famous quote of George Box that “all models are wrong but some are useful” [9]. Although some use that quote to motivate carelessness model building, Box was arguing the opposite. Only with respect for the given



application and careful work can we develop an observational model that contains model configurations that sufficiently

well approximate the true data generating process, providing the foundation for useful inferences.

## 1 Probabilistic Modeling

Principled model development strives for an observational process that captures the relevant structure of the phenomenon, environment, and probe that give rise to the true data generating process. We always have to be careful, to recognize the

limitations of the even the most carefully designed observational model in any given analysis.

### 1.3 The Observational Model

## 1.3.1 The Generative Structure of an Observational Model

### 1.4 Model-Based Inferences

A particularly effective way to build useful observational models is to design each model configuration to emulate the

## 2 Frequentist Inference

generative structure of the observational process, from the latent phenomenon through the surrounding environment to the

final observational probe. The parameters of a model following the generative structure of the observational process

## 3 Bayesian Inference

typically decompose into parameters for each layer,

## 4 Comparing Asymptotic Apples to

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$$\theta = \{\theta_{\text{phenom}}, \theta_{\text{envir}}, \theta_{\text{probe}}\}.$$

## 5 Conclusion

Keep in mind that the separation between environmental and probe parameters, need not always be so clean, especially for

cruder approximations to the true data generating process. For example is the influence of the person collecting the data

better classified as a part of the environment or a part of the probe? Taken together the non-phenomenological parameters

are sometimes denoted *systematic parameters*.

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In the Gaussian example introduced above we might consider the location parameter,  $\mu$ , as a phenomenological parameter that quantifies some baseline effect. From this perspective the scale parameter,  $\sigma$ , would capture more systematic structure relating to either the surrounding environment or the particulars of the experimental probe. Another example where this decomposition is natural is naive curve fitting, where the curve can be interpreted as implicitly modeling the behavior of the latent phenomenon while the residuals around that curve model the rest of the observational process.

In addition to facilitating the design of the observational model for a single probe, this generative decomposition also motivates principled observational models for multiple probes. For example two probes might interrogate the same phenomenon without experiencing exactly the same environment and probe circumstances. Consequently we might specify the observational model as

$$\begin{aligned} \pi_{\square}(y; \theta) = & \pi_{\square}(y_1; \theta_{\text{phenom}}, \theta_{\text{envir}}, \theta_{\text{envir},1}, \theta_{\text{probe}}, \theta_{\text{probe},1}) \\ & \cdot \pi_{\square}(y_2; \theta_{\text{phenom}}, \theta_{\text{envir}}, \theta_{\text{envir},2}, \theta_{\text{probe}}, \theta_{\text{probe},2}), \end{aligned}$$

where the subscripted parameters, such as  $\theta_{\text{envir},1}$  are unique to a specific context and the non-subscripted parameters, such as  $\theta_{\text{envir}}$ , are shared across contexts. The better we can model the systematic structure of the observational process the better we can build analyses that study the same phenomenon across different experiments, or develop hypotheses about the outcomes of new experiments targeting the same phenomenon.

Consider, for example, a latent phenomenon that influences a binary response in the environment. We might model the phenomenological behavior with a probability,  $p$ , of the binary response taking the value  $y = 1$  instead of  $y = 0$ . The rest of the observational model depends on the particular structure of the observational probe.

If we probe the system once and record the binary outcome then we might utilize an observational model consisting of Bernoulli distributions parameterized by the phenomenological probability,

$$\pi_{\square}(y; p) = \text{Ber}(y \mid p).$$

Presuming that the system is memoryless, repeated probes might be modeled as a product of independent Bernoulli distributions that are all parameterized by the same phenomenological probability,

$$\pi_{\square}(y_1, \dots, y_N; p) = \prod_{n=1}^N \text{Ber}(y_n \mid p).$$

Alternatively we might probe the system  $N$  times but only record the total number of  $y = 1$  responses,  $k$ . In this case we could model the observational process with binomial distributions parameterized with the phenomenological probability,

#### 1.1 The Observational Process

$$\pi_{\square}(k; p) = \text{Bin}(k \mid p).$$

Finally, probing until  $k$  binary responses are observed could be modeled with negative Binomial distributions parameterized by the phenomenological probability,

#### 1.3 The Observational Model

#### 1.4 Model-Based Inferences

$$\pi_{\square}(k; p) = \text{NegBin}(k \mid p).$$

A simple binary response quantified by a phenomenological probability, however, is a relatively crude approximation to any realistic observational process. The more deeply we resolve the observational process the richer our observational model will be as an inference.

The binary response for Apple might correspond to the capture of a particular species in an ecological experiment. In this case the probability of capture might not be a single, monolithic parameter but rather the outcome of the population dynamics between the species of interest, the species on which it preys, and the species which predate it.

It could also correspond to the alarm of a detector that has recorded enough incident radiation to pass a certain threshold. Here the latent probability would be a consequence of the possible sources of radiation, including radiation from particular sources of interest as well as the irrelevant background sources that permeate the environment. Depending on the precision required of the desired inferences we might also have to model the detector response itself with additional parameters, including those quantifying the calibration of the incident radiation flux and the behavior of the electronics Original Content Integrated to trigger the detection.

Each deeper look into the observational process discerns more detail and a richer generative structure. At the same time, in a given application we might need only a crude approximation that blends together various aspects of the generative structure. Perhaps the biggest challenge in model building is determining exactly how sophisticated an observational model needs to be in order to achieve the desired inferential goals.

## 1.3.2 Limitations of Observational Models

Regardless of how sophisticated our observational model might be, it will always be an approximation to the complexity of the true data generating process. We have to be ever mindful of the limitations of the observational model in the context of any particular analysis.

For example, the observational model might approximate the true data generating process only within a certain regime of behaviors. At the same time the approximations might accurately capture only some behaviors but not others. A given observational model can express the large scale structure of the latent system but make no attempt to incorporate the structure at smaller scales. It might capture statistical relationships between phenomenological parameters but fail to accurately convey the physical dynamics or explicit causal structure of the phenomenon and its interaction with the environment.

Consequently we have to be extremely careful to employ model-based inferences only within the regimes where the approximations are reasonable and sufficient for the decisions that we want to inform. In other words, we have to develop a model that is sophisticated enough for the given application.

Another subtlety with the approximate nature of the observational model is that a given observational model might be a reasonable approximation for *many* true data generating processes. This is particularly common for observational models that focus on capturing the *emergent* behaviors of a complex observational process. At sufficiently coarse scales much of the latent structure washes out and only the most coherent of the latent behaviors survive. All true data generating process with similar emergent behavior can then be equally well approximated by the same observational model.

Linear regression, for example, can be interpreted as a model where phenomenological behavior manifests as a deterministic linear relationship between observed variates and covariates which is then convolved through Gaussian fluctuations. This linear relationship, however, can be considered as a first-order approximation to *any* latent

phenomenological behavior. At the same time Gaussian variation well approximates a wide range of environmental and probe behaviors, especially within the context of the first-order behavior. Consequently any true data generating process with the same first-order phenomenological behavior, and emergent environmental and probe behaviors, will be reasonably approximated by the *same* linear regression model.

Emergent models cannot discriminate between the true data generating processes they approximate and they cannot express behaviors outside of the scope of the approximation. The meaningful interpretation of a model-based statistical analysis requires not just an explicit observational model but also the scope of the approximations it presumes. In other words, we need to at least conceptually understand the context of the model relative to the true data generating process.

### 1.3.3 Model-Based Calibration

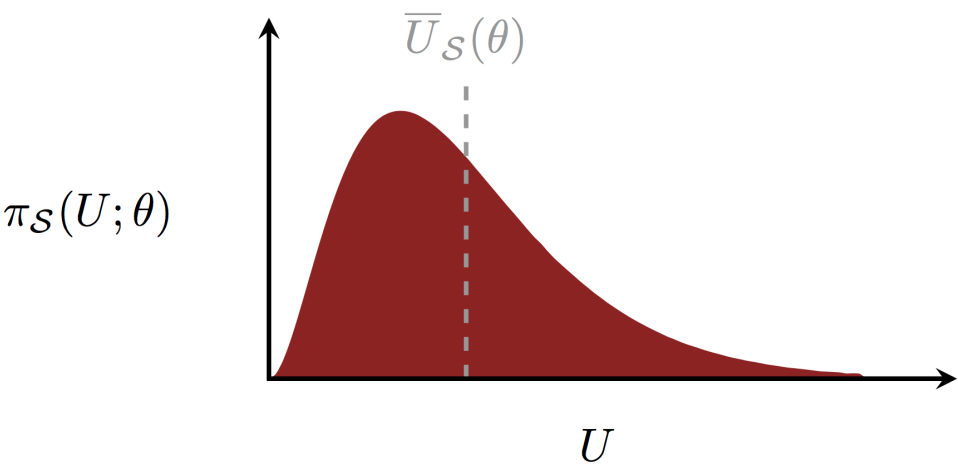
While we cannot perfectly calibrate a decision-making process without knowing the true data generating process, we can define a proxy calibration within the context of a given observational model.

Firstly we assume that we can define utilities for all available actions *within the context of each model configuration*,

$$U : A \times \Theta \rightarrow \mathbb{R}^+ \\ (a, \theta) \mapsto U(a, \theta).$$

$$U \circ a : Y \times \Theta \rightarrow \mathbb{R}^+ \\ (y, \theta) \mapsto U(a(y), \theta).$$

We can push each of the data generating processes in our observational model through the composite utility function to give a full distribution of possible utilities for each model configuration,  $\pi_{\square}(U; \theta)$ .



The properties of this distribution then provide the foundation to define various calibrations. For example, for each model configuration we can quantify the *expected utilities*,

$$\overline{U}_{\square}(\theta) = \int \mathrm{d}y \, \pi_{\square}(y; \theta) \, U(a(y), \theta)].$$

How we might exploit these calibrations in an analysis will then depend on how we handle their dependence on the unknown model configuration. No matter how we end up employing these model-based utilities we have to be vigilant in recognizing they are only as good as the observational model itself.

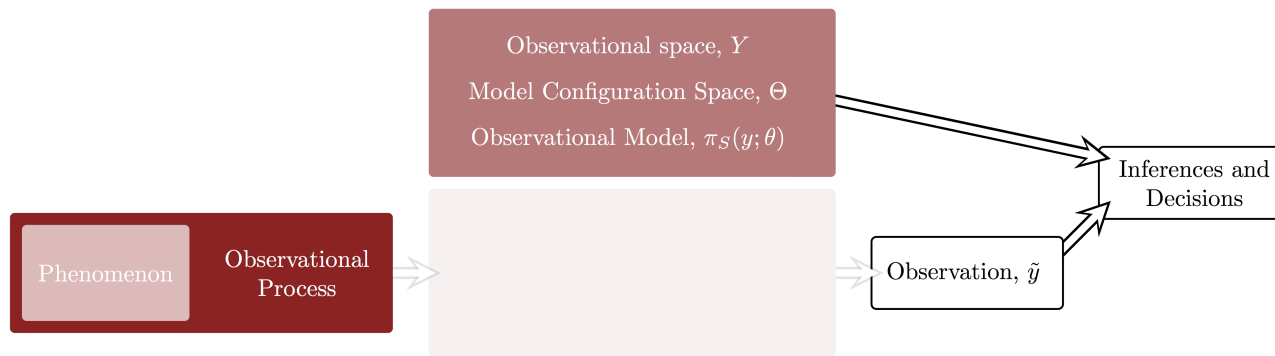
#### 1.1 The Observational Process

## 1.4 Model-Based Inferences

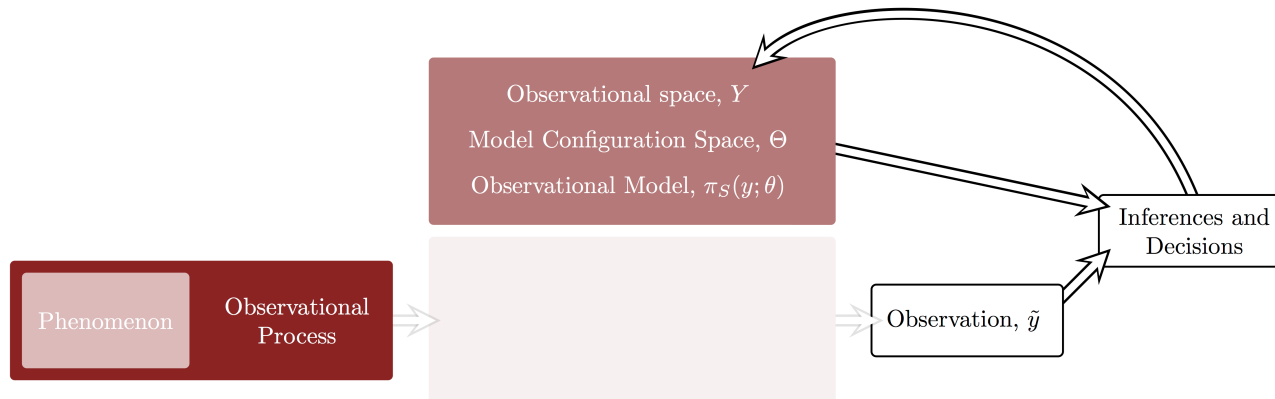
Once an Observational Model has been established, and its limitations acknowledged, we can consider *model-based inferences* that identify which model configurations are most consistent with the true data generating process. Unfortunately without knowing the true data generating process these inferences can be informed only by the information inherent in observed data and our own domain expertise. Within the context of an observational model, inference becomes the identification of how *consistent* model configurations are with the observed data and any domain expertise we introduce.

Consequently the observational model both plays two critical roles. Firstly it defines a context for constructing inferences from observations.

#### 5 Conclusion



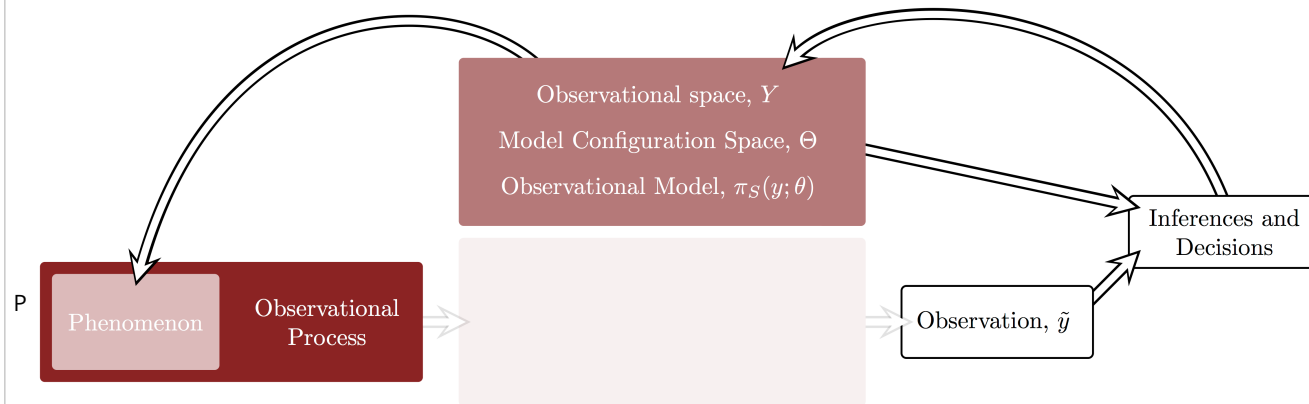
In particular, model-based inferences take the form of a quantification of which model configurations are consistent with the observed data.



Secondly those model-based inferences then inform decisions about how to interact with the latent phenomenon, bridging the gap between the observed data and the latent system.

## 1 Probabilistic Modeling

### 1.1 The Observational Process



## References

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The utility of those decisions will of course depend on how well the observational model approximates the relevant properties of the true data generating process.

In order to formally define inference we to define how consistency is mathematically encoded, and then how to calculate consistency assignments in light of a given observation. Unfortunately there is no unique way to encode consistency, and hence no unique way of constructing inferences.

# 2 Frequentist Inference

The methodology most common to introductory statistics courses is the *frequentist* approach which introduces additional philosophical assumptions to constrain the possible quantifications of consistency.

Frequentist methods ultimately derive from attempts to make probability *objective* by defining it as something which could be measured. More formally, this approach assumes that probabilities are *defined* as the long run frequencies of events that can be repeated infinitely often. For example the probability of rolling a given face on a die is defined as the ratio of rolls that yield that face over the total number of rolls, as the die rolling is repeated an infinite number of times. Because all observers would hypothetically encounter the same limiting frequencies, these probabilities would be physical and hence independent of the observer.

Unfortunately this definition quickly runs into technical difficulties [10]. Firstly, we can never actually realize the infinite, perfect replications needed to define the limiting frequencies in practice. An infinite number of rolls would not only take infinitely long to perform, they would also take their toll on the integrity of the die and the arm of the poor soul entrusted with the Sisyphean task. Moreover, even if we ignore the infeasibility of the replications we cannot ignore the mathematical difficulties with defining limiting frequencies. It turns out that even in simple cases these limits are not always so well-behaved.

Consequently formal frequentist inference pulls something of a bait and switch – the frequency definition of probabilities is replaced with Kolmogorov's axioms, but with the restriction that the resulting probability theory can be applied to *only* events that can be repeated infinitely often, at least hypothetically. The hypothetical nature of this restriction has some interesting loopholes that one sometimes has to exploit to model populations and other systems of interest, but it does provide a mathematical foundation on which we can build inferences. In particular, if we presume that the circumstances of an observation can be exactly replicated then we can use probability theory to describe the distribution of possible observations and admit the observational models that we introduced above.

The only way that we can define consistency under these assumptions, however, is through *binary quantifications*. Individual model configurations are either consistent with the observation or they're not. Any continuous, and mathematically self-consistent, quantification of consistency would be equivalent to the probability theory which is forbidden from consorting with the model configuration space.

An immediate weakness of binary configurations is that once a model configuration is deemed inconsistent it can no longer inform decisions. An overaggressive quantification can be lead astray by irrelevant features of an observation and end up discarding model configurations that well approximate the true data generating process, if not the true data generating process itself. To compensate for this frailty the frequentist approach treats inference as a *decision-making process* and carefully calibrates those decisions to ensure that the binary quantifications are well-behaved *on average*.

Formally, consistency is assigned by *estimators* that deterministically map observations into some binary quantifications of the model configuration space. If we treat the possible quantifications as actions,  $a$ , then an estimator defines a decision-making process,  $a(\cdot)$ , which applies to

Preasymptotic Oranges  
If we presume that some  $\theta$  is the true model configuration then we can use the corresponding data generating process to define the expected utility of a given estimator,

Acknowledgements

References

The remaining challenge is accounting for the dependence on the presumed model configuration,  $\theta$ .

Because of the foundational assumptions of the frequentist perspective, however, we are limited here. For example we cannot average over the possible model configurations because as that would require applying probability theory to the model configuration space. Just about the only thing that we can do is report the *worst case expected utility*,

$$U^* = \operatorname{argmin}_{\theta \in \Theta} \overline{U}(\theta).$$

In this way we can calibrate the worst case behavior of the estimator and determine whether or not it meets our needs, at least under the assumptions that the true data generating process is contained within the observational model.

In the end frequentist inference does not define a unique inferential procedure. Instead it formalizes inferential procedures as estimators and defines a unique procedure for calibrating those estimators. In order to develop a frequentist analysis we need to specify not just our observational model but also an estimator and its associated utility function, and then carry out the worst case expected utility analysis.

To demonstrate frequentist methods let's consider two classes of estimators, *point estimators* and *set estimators*, and their explicit calibration in a simple example before remarking on what they teach us about implementing frequentist methods in practice. Those uninterested in the machinery of frequentist inference can freely skip these sections and continue on to the following introduction of Bayesian inference.

## 2.1 Point Estimators

A *point estimator* maps a given observation into single model configuration,

$$\begin{aligned} \hat{\theta}: Y &\rightarrow \Theta \\ y &\mapsto \hat{\theta}(y). \end{aligned}$$

A natural utility function that complements a point estimator is the squared error,

## 1 Probabilistic Modeling

The more accurate and precise our point estimator is, the smaller the squared error will be and the more utility the estimator will be assigned.

### 1.2 The True Data Generating Process

For each model configuration we'd then compute the expected utility

### 1.3 The Observational Model

### 1.4 Model-Based Inferences

## 2 Frequentist Inference

and summarize the expected performance of the point estimator with the worst case expected utility.

## 3 Bayesian Inference

For example, consider a demonstrative observational model that consists of  $N$  independent Gaussian probability density functions,

Preasymptotic Oranges

## 5 Conclusion

## Acknowledgements

Before the empirical mean,

License

Original Computing Environment

$$U : Y \times \Theta \rightarrow -\mathbb{R}^+$$

$$(y, \theta) \mapsto -(\hat{\theta}(y) - \theta)^2.$$

$$\overline{U}(\theta) = - \int dy \pi_{\square}(y; \theta) (\hat{\theta}(y) - \theta)^2.$$

$$\pi(y; \theta) = \prod_{n=1}^N \square(y_n; \mu, \sigma) = \prod_{n=1}^N \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{y_n - \mu}{\sigma}\right)^2\right)$$

$$\hat{\mu}(y) = \frac{1}{N} \sum_{n=1}^N y_n.$$

as a point estimator for the location parameter,  $\mu$ , is particularly useful because it features a conveniently analytic pushforward probability density function,

$$\pi(\hat{\mu}; \mu, \sigma) = \square\left(\hat{\mu}; \mu, \frac{\sigma}{\sqrt{N}}\right).$$

In terms of the estimator the squared error utility function becomes

$$U(\hat{\mu}; \mu, \sigma) = -(\hat{\mu} - \mu)^2.$$

The worst possible utility stretches all the way towards negative infinitely, corresponding to the estimator being infinitely far away from the true value of the location parameter. On the other hand the best possible utility is zero, corresponding to the estimator exactly recovering the true value. The expected utility with respect to our observational model becomes

$$\overline{U}(\mu, \sigma) = - \int_{-\infty}^{\infty} d\mu \pi\left(\hat{\mu}; \mu, \frac{\sigma}{\sqrt{N}}\right) (\hat{\mu} - \mu)^2$$

which gives another analytic result,

$$\overline{U}(\mu, \sigma) = -\frac{\sigma^2}{N}.$$

If  $\sigma$  is known then the expected utility of the estimator is ,

$$\overline{U}(\mu) = -\frac{\sigma^2}{N}$$

and the worst case loss becomes

$$\overline{U}^* = -\frac{\sigma^2}{N}.$$

This independence of the expected utility with respect to the model configurations is quite exceptional and depends on the serendipitous interactions between this particular point estimator, utility function, and Gaussian observational model.

1 Probabilistic Modeling  
When  $\sigma$  is not known then the worst case utility has to consider the edge of the model configuration space where  $\sigma$  approaches infinity and, for a fixed observation size  $N$ , the utility becomes arbitrarily poor. The only way to ensure a constrained worst case expected utility is to confine  $\sigma$  below some upper limit.

1.2 The True Data Generating Process  
These calculations are prone to mathematical errors for even the most experienced statistician. We can also consider Monte Carlo estimation of the expected utilities, however, to verify the analytic results, if not outright replace them. First let's set up our 4 Monte Carlo inference functions that compute Monte Carlo estimators and their standard errors.

```
2 Frequentist Inference
c_tight <- c("#DCBCBC")
3 Bayesian Inference
c_loose <- c("#8E2727")
4 Comparing Asymptotic Apples to
Preasymptotic Oranges
welford_summary <- function(x) {
5 Conclusion
summary = c(0, 0)
6 Acknowledgement
for (i in 1:length(x)) {
delta <- x[i] - summary[1]
7 References
summary[1] <- summary[1] + delta / (n + 1)
summary[2] <- summary[2] + delta * (x[i] - summary[1])
8 License
}
9 Original Computing Summary Est / (length(x) - 1)
return(summary)
}

compute_mc_stats <- function(x) {
summary <- welford_summary(x)
return(c(summary[1], sqrt(summary[2] / length(x))))
}
```

We'll assume 10 observations and use 1000 simulations across a grid of possible data generating processes.

```
N_obs <- 10

N_simu <- 1000

N_mu <- 3
mus <- seq(-3, 3, 3)

N_sigma <- 20
sigmas <- seq(0.5, 10, 0.5)
```

We then define our utility function and location estimator,

```
se_util <- function(x_hat, x) {
-(x_hat - x)**2
}

mu_hat <- function(y) {
mean(y)
}
```

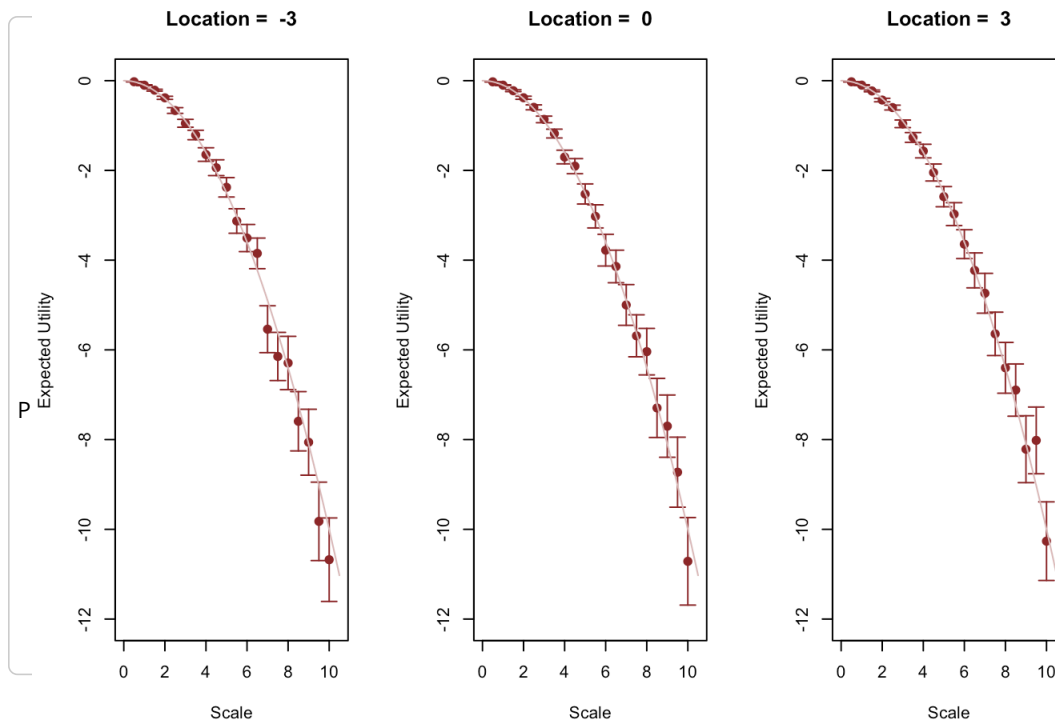


and turn the crank,

```
1 Probabilistic Modeling
par(mfrow=c(1, 3))
1.1 The Observational Process
for(m in 1:N_mu) {
1.2 The True Data Generating Process
  exp_utils[1, m] <- rep(0, N_sigma)
1.3 The Observational Model
  deltas <- rep(0, N_sigma)
1.4 Models Based in Noise {
    sample_util <- rep(0, N_simu)
2 Frequentist Inference
  for (s in 1:N_simu) {
3 Bayesian Inference
    simu_data <- rnorm(N_obs, mus[m], sigmas[s])
4 Comparing Asymptotic Apple to mu_hat(simu_data)
    mu_hat[s] <- mu_hat(simu_data)
Preasymptotic Orange[n] <- se_util(location_estimator, mus[m]);
  }
5 Conclusion
  mc_stats <- compute_mc_stats(sample_util)
  exp_utils[s] <- mc_stats[1]
Acknowledgements
  deltas[s] <- 2 * mc_stats[2]
References
License
  plot(sigmas, exp_utils, pch=19, col=c_dark, main=paste("Location = ", mus[m]),
Original Colored Environment
  ylim=c(-12, 0), ylab="Expected Utility")

  arrows(sigmas, exp_utils - deltas, sigmas, exp_utils + deltas,
    length=0.05, angle=90, code=3, col=c_dark)

  xs <- seq(0, 10.5, 0.01)
  lines(xs, -xs * xs / N_obs, col=c_light)
}
```



What about the scale parameter,  $\sigma$ ? With the guidance of our elders we might consider the empirical standard deviation,

$$\sigma(\hat{y}) = \sqrt{\frac{1}{N-1}(\mu(\hat{y}) - y_n)^2},$$

as a point estimator for  $\sigma$ . This point estimator once again enjoys a analytic pushforward probability density function,

$$\pi(\sigma) = \chi\left(\sigma; N-1, \frac{\sigma}{\sqrt{N-1}}\right).$$

Here  $\chi$  refers to the scaled- $\chi$  probability density function with degrees of freedom,  $\nu$ , and scale,  $s$ ,

$$\chi(y; \nu, s) = s^{-\nu} \frac{2^{1-\frac{\nu}{2}}}{\Gamma(\frac{\nu}{2})} y^{\nu-1} e^{-\frac{1}{2} \frac{y^2}{s^2}}$$

The expected squared error utility is then given by

$$\overline{U}(\mu, \sigma) = - \int_{-\infty}^{\infty} d\sigma \chi\left(\sigma; N-1, \frac{\sigma}{\sqrt{N-1}}\right) (\sigma - \sigma)^2$$

which, after some tedious algebraic torture, surrenders the result

$$\overline{U}(\mu, \sigma) = -2 \left( 1 - \sqrt{\frac{2}{N-1} \frac{\Gamma(\frac{N}{2})}{\Gamma(\frac{N-1}{2})}} \right) \sigma^2$$

Like the location estimator when  $\sigma$  isn't known, the worst case expected squared error utility is arbitrarily bad for fixed  $N$

unless  $\sigma$  can be exactly confined below some upper limit. From a strict frequentist perspective we can't trust these estimators unless we can upper bound the true value of the scale parameter and avoid that extreme of the model configuration space.

1.1 The Observational Process  
As before we can readily compare these results to Monte Carlo calculations.

1.2 The True Data Generating Process

sigma\_hat <- function(y) {

1.3 The Observational Model

sd(y)

}  
1.4 Model-Based Inferences

2 Frequentist Inference

3 Bayesian Inference

for(m in 1:N\_mu) {

4 Computing Asymptotic Approximations

exp\_utils <- rep(0, N\_sigma)

deltas <- rep(0, N\_sigma)

5 Conclusion

for(s in 1:N\_sigma) {

sample\_util <- rep(0, N\_simu)

Acknowledgements

References

for(n in 1:N\_simu) {

simu\_data <- rnorm(N\_obs, mus[m], sigmas[s])

scale\_estimator <- sigma\_hat(simu\_data)

sample\_util[n] <- se\_util(scale\_estimator, sigmas[s]);

}  
Original Computing Environment

mc\_stats <- compute\_mc\_stats(sample\_util)

exp\_utils[s] <- mc\_stats[1]

deltas[s] <- 2 \* mc\_stats[2]

}  
plot(sigmas, exp\_utils, pch=19, col=c\_dark, main=paste("Location = ", mus[m]),

xlim=c(0, 10.5), xlab="Scale",

ylim=c(-7, 0), ylab="Expected Utility")

arrows(sigmas, exp\_utils - deltas, sigmas, exp\_utils + deltas,

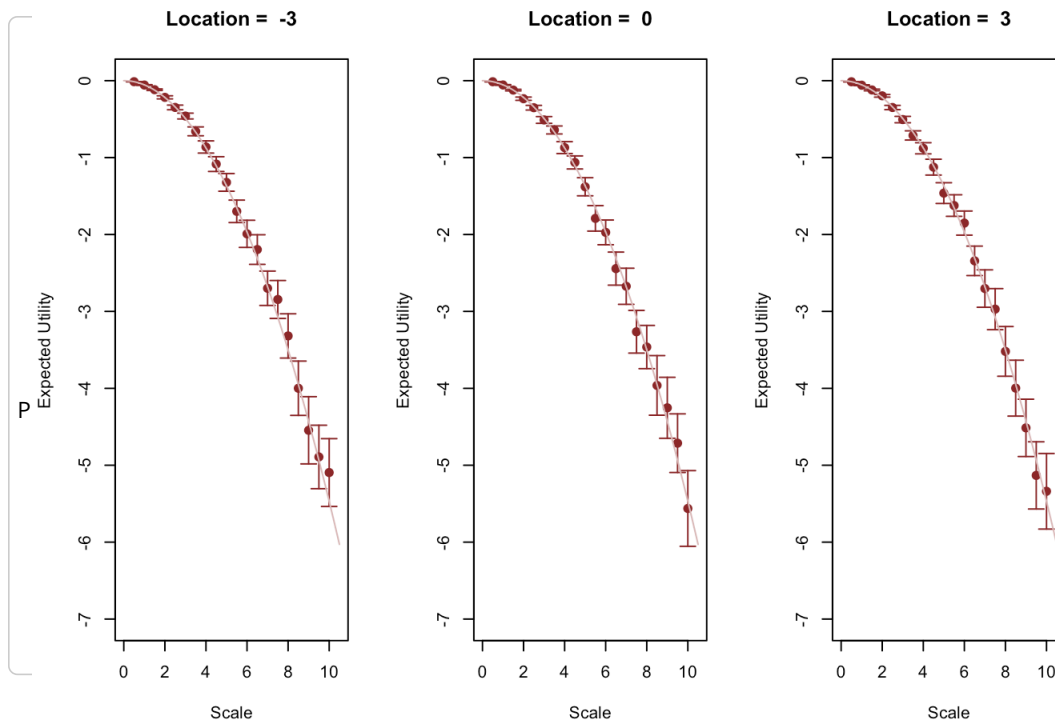
length=0.05, angle=90, code=3, col=c\_dark)

C\_N <- 2 \* (1 - sqrt(2 / (N\_obs - 1)) \* gamma(0.5 \* N\_obs) / gamma(0.5 \* (N\_obs - 1)))

xs <- seq(0, 10.5, 0.01)

lines(xs, -C\_N \* xs \* xs, col=c\_light)

}



## 2.2 Set Estimators

Alternatively we might consider *set estimators*, also known as confidence sets or confidence intervals, that map observations into entire sets of model configurations,

$$\begin{aligned}\widehat{\Delta\theta} : Y &\rightarrow \square \\ y &\mapsto \widehat{\Delta\theta}(y),\end{aligned}$$

where  $\square$  is an appropriate  $\sigma$ -algebra over the model configuration space. In this case a natural utility function is the *inclusion utility function* that considers whether or not the estimated set includes the presumed true model configuration,

$$\begin{aligned}U : Y \times \Theta &\rightarrow \{0, 1\} \\ (y, \theta) &\mapsto \mathbb{I}[\theta \in \widehat{\Delta\theta}(y)].\end{aligned}$$

The expected utility for a given model configuration is then the probability of inclusion, or *coverage*,

$$\begin{aligned}\overline{U}(\theta) &= \int dy \pi_{\square}(y; \theta) \mathbb{I}[\theta \in \widehat{\Delta\theta}(y)] \\ &= \mathbb{P}[\theta \in \widehat{\Delta\theta}(y)].\end{aligned}$$

In other words, our set estimator is playing ring toss around the presumed truth (<https://twitter.com/epiellie/status/1073385427317465089>) and we want to know how often we score a ringer. The final

calibration of a confidence set is then the smallest coverage across all possible model configurations,

## 1 Probabilistic Modeling

### 1.1 The Observational Process

If  $\sigma$  is known then we can readily construct a set estimator for the location parameter as an interval centered around the location point estimator we considered above,

### 1.3 The Observational Model

### 1.4 Model-Based Inferences

To facilitate the computation of the expected inclusion utility we begin by writing the inclusion utility function in terms of the point estimator for the location. First we split the indicator function for the entire set estimator into separate indicator functions for each end of the interval estimator,

## 4 Comparing Asymptotic Apples to Preasymptotic Oranges

## 5 Conclusion

**Known objects:** the definitions for each boundary and solve for the equivalent utility as a function of the location estimator,  $\mu(y)$ ,  
References

## License

## Original Computing Environment

$$U^* = \operatorname{argmin}_{\theta \in \Theta} \mathbb{P}[\theta \in \widehat{\Delta\theta}(y)].$$

$$\widehat{\Delta\mu}(y) = [\mu_1^{\wedge}(y), \mu_2^{\wedge}(y)] = [\mu(y) - \delta \cdot \sigma, \mu(y) + \delta \cdot \sigma].$$

$$\begin{aligned} U(\widehat{\Delta\mu}(y); \mu, \sigma) &= \mathbb{I}[\mu_1^{\wedge}(y) \leq \mu \leq \mu_2^{\wedge}(y)] \\ &= \mathbb{I}[\mu_1^{\wedge}(y) \leq \mu] \cdot \mathbb{I}[\mu \leq \mu_2^{\wedge}(y)]. \end{aligned}$$

$$\begin{aligned} U(\mu(y); \mu, \sigma) &= \mathbb{I}[\mu_1^{\wedge}(y) \leq \mu] \cdot \mathbb{I}[\mu \leq \mu_2^{\wedge}(y)] \\ &= \mathbb{I}[\mu(y) - \delta \cdot \sigma \leq \mu] \cdot \mathbb{I}[\mu \leq \mu(y) + \delta \cdot \sigma] \\ &= \mathbb{I}[\mu(y) \leq \mu + \delta \cdot \sigma] \cdot \mathbb{I}[\mu(y) > \mu - \delta \cdot \sigma]. \end{aligned}$$

We can now use the known pushforward distribution for the location point estimator to compute the expected inclusion utility,

$$\begin{aligned} \overline{U}(\mu) &= \int_{-\infty}^{\infty} d\mu \square\left(\mu; \mu, \frac{\sigma}{\sqrt{N}}\right) \mathbb{I}[\mu(y) \leq \mu + \delta \cdot \sigma] \cdot \mathbb{I}[\mu(y) \geq \mu - \delta \cdot \sigma] \\ &= \int_{\mu - \delta \cdot \sigma}^{\mu + \delta \cdot \sigma} d\mu \square\left(\mu; \mu, \frac{\sigma}{\sqrt{N}}\right) \\ &= \Phi\left(\mu + \delta \cdot \sigma; \mu, \frac{\sigma}{\sqrt{N}}\right) - \Phi\left(\mu - \delta \cdot \sigma; \mu, \frac{\sigma}{\sqrt{N}}\right), \end{aligned}$$

where  $\Phi$  is the Gaussian cumulative distribution function. Taking advantage of the fact that

$$\Phi(x; \mu, \sigma) = \Phi\left(\frac{x - \mu}{\sigma}; 0, 1\right)$$

we can reduce the expected utility even further to

$$\begin{aligned} \overline{U}(\mu) &= \Phi\left((\mu + \delta \cdot \sigma - \mu) \frac{\sqrt{N}}{\sigma}; 0, 1\right) - \Phi\left((\mu - \delta \cdot \sigma - \mu) \frac{\sqrt{N}}{\sigma}; 0, 1\right) \\ &= \Phi\left((\delta \cdot \sigma) \frac{\sqrt{N}}{\sigma}; 0, 1\right) - \Phi\left(-(\delta \cdot \sigma) \frac{\sqrt{N}}{\sigma}; 0, 1\right) \\ &= \Phi\left(\delta \cdot \sqrt{N}; 0, 1\right) - \Phi\left(-\delta \cdot \sqrt{N}; 0, 1\right). \end{aligned}$$

Finally, noting that

$$\Phi(-x; \mu, \sigma) = 1 - \Phi(x; \mu, \sigma)$$

## 1 Probabilistic Modeling

### 1.1 The Observational Process

### 1.2 The True Data Generating Process

The expected utility is once again independent of the true value of  $\mu$ , so the worst case expected utility is simply

### 1.3 The Observational Model

### 1.4 Model-Based Inferences

Note that this worst case expected utility increases towards zero as  $N$  increases; the interval estimator becomes more and more accurate as the size of the observation increases.

## 3 Bayesian Inference

In many introductory statistics textbooks one commonly finds the choice  $\delta = z_{1-\alpha/2}/\sqrt{N}$  for a given  $\alpha$ , where  $z_q$  is the

standard normal quantile function. This choice ensures that the coverage of the interval estimator is exactly  $1 - \alpha$ .

## 4 Comparing Asymptotic Apples to

Preasymptotic Oranges  
Tired yet? Well too bad because we still have to consider a set estimator for  $\mu$  when  $\sigma$  is not known. An immediate

approach is to take the interval estimator we used above and simply replace  $\sigma$  with its point estimator,

## Acknowledgements

$$\widehat{\Delta\mu}(y) = [\mu_1^{\wedge}(y), \mu_2^{\wedge}(y)] = [\mu(y) - \delta \cdot \sigma(y), \mu(y) + \delta \cdot \sigma(y)].$$

## References

The immediate challenge with this interval estimator is that the pushforward distributions for  $\widehat{\Delta\mu}(y)$  cannot be computed from the marginal pushforward distributions for  $\mu(y)$  and  $\sigma(y)$  because the distributions are not independent.

## Original Computing Environment

We don't, however, have to compute the full pushforward distribution  $\widehat{\Delta\mu}(y)$ . If we manipulate the inclusion utility function to isolate the point estimators,

$$\begin{aligned} U(I(\hat{y}); \mu, \sigma) &= \mathbb{I}[\mu(y) - \delta \cdot \sigma(y) \leq \mu] \cdot \mathbb{I}[\mu \leq \mu(y) + \delta \cdot \sigma(y)] \\ &= \mathbb{I}\left[\frac{\mu(y) - \mu}{\sigma(y)} \leq \delta\right] \cdot \mathbb{I}\left[\frac{\mu(y) - \mu}{\sigma(y)} > -\delta\right] \end{aligned}$$

then we see that the inclusion utility function for this interval estimator depends on the observed data only through the ratio

$$\rho(y) = \frac{\mu(y) - \mu}{\sigma(y)}.$$

In other words,

$$U(\widehat{\Delta\mu}(y); \mu, \sigma) = \mathbb{I}[\rho(y) \leq \delta] \cdot \mathbb{I}[\rho(y) > -\delta].$$

This is a significant improvement because the pushforward distribution for  $\rho(y)$  can be worked out analytically to be specified by a Student-t probability density function with  $N - 1$  degrees of freedom,

$$\pi(\rho(y)) = t\left(\rho(y); N - 1, 0, \frac{1}{\sqrt{N}}\right).$$

We can now compute the expected inclusion utility as

$$\begin{aligned} \overline{U}(\mu, \sigma) &= \int_{-\infty}^{\infty} d\rho \gamma\left(\rho(y); N-1, 0, \frac{1}{\sqrt{N}}\right) \mathbb{I}[\rho(y) \leq \delta] \cdot \mathbb{I}[\rho(y) > -\delta] \\ 1.1 \text{ The Observational Process} & \\ 1.2 \text{ The True Data Generating Process} &= \int_{-\delta}^{\delta} d\rho \gamma\left(\rho(y); N-1, 0, \frac{1}{\sqrt{N}}\right) \\ 1.3 \text{ The Observational Model} & \\ 1.4 \text{ Model-Based Inferences} &= T\left(\delta; N-1, 0, \frac{1}{\sqrt{N}}\right) + -T\left(\delta; N-1, 0, \frac{1}{\sqrt{N}}\right) \\ 2 \text{ Frequentist Inference} &= 2 T\left(\delta; N-1, 0, \frac{1}{\sqrt{N}}\right) - 1, \\ 3 \text{ Bayesian Inference} & \\ 4 \text{ Comparing Asymptotic Apples to Preasymptotic Oranges} &= 2 T\left(\delta * \sqrt{N}; N-1, 0, 1\right) - 1, \end{aligned}$$

where  $T$  is the Student-t cumulative distribution function. As in the case of known scale,  $\sigma$ , the expected utility does not depend on either the location or scale and so the worst case expected utility immediately becomes

$$\overline{U}^* = 2 T\left(\delta * \sqrt{N}; N-1, 0, 1\right) - 1.$$

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As with the point estimators, the very delicate cooperation between the chosen set estimator, the inclusion utility function, and the Gaussian observational model yields analytic results all the way through, at least if we know exactly which path to take through the calculations. At least we can verify these results by comparing them to Monte Carlo calculations.

```

inc_util <- function(x_hat_left, x_hat_right, x) {
  x_hat_left <- x & x <= x_hat_right
}
1 Probabilistic Modeling
}
1.1 The Observational Process
mu_hat_left <- function(y, delta) {
1.2 The True Data Generating Process
  mu_hat(y) - delta * sigma_hat(y)
}
1.3 The Observational Model
1.4 Model-Based Inferences
mu_hat_right <- function(y, delta) {
2 Frequentist Inference
  mu_hat(y) + delta * sigma_hat(y)
}
3 Bayesian Inference
delta_interval <- 0.5
4 Comparing Asymptotic Apples to
Preasymptotic Oranges
par(mfrow=c(1,3))
5 Conclusion
for(m in 1:N_mu) {
Acknowledgements rep(0, N_sigma)
deltas <- rep(0, N_sigma)
References
for(s in 1:N_sigma) {
License
  sample_util <- rep(0, N_simu)
Original Computing Environment

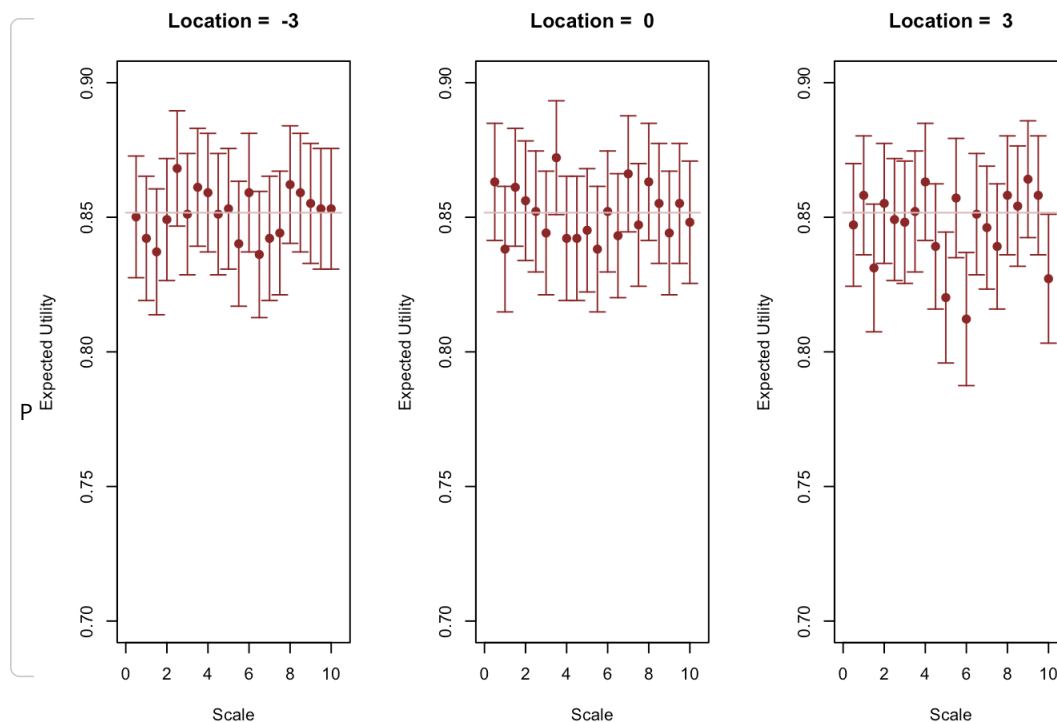
  for (n in 1:N_simu) {
    simu_data <- rnorm(N_obs, mus[m], sigmas[s])
    left_est <- mu_hat_left(simu_data, delta_interval)
    right_est <- mu_hat_right(simu_data, delta_interval)
    sample_util[n] <- inc_util(left_est, right_est, mus[m]);
  }
  mc_stats <- compute_mc_stats(sample_util)
  exp_utils[s] <- mc_stats[1]
  deltas[s] <- 2 * mc_stats[2]
}
plot(sigmas, exp_utils, pch=19, col=c_dark, main=paste("Location = ", mus[m]),
     xlim=c(0, 10.5), xlab="Scale",
     ylim=c(0.7, 0.9), ylab="Expected Utility")

arrows(sigmas, exp_utils - deltas, sigmas, exp_utils + deltas,
       length=0.05, angle=90, code=3, col=c_dark)

xs <- seq(0, 10.5, 0.01)
ys <- rep(2 * pt(delta_interval * sqrt(N_obs), N_obs - 1) - 1, length(xs))
lines(xs, ys, col=c_light)
}

```





## 2.3 Frequentist Inference in Practice

As we saw above, the worst case calibration used by frequentist methods can be very conservative. An estimator might be well-behaved for most reasonable model configurations but have its worst case behavior determined by the boundaries of the model configuration space populated by extreme, but not impossible, model configurations. That said, if we are fortunate enough to engineer an estimator with an acceptable worst case expected utility then we can have confidence that its behavior will be extremely robust, at least within the context of the observational model. The practical challenge is whether or not such an estimator even exists and we can find it.

Even if such an estimator does exist, however, performing the proper frequentist calibration needed to determine the worst case expected utility isn't always feasible. Despite the simplicity of the observational model in the examples shown above, for example, the calculations were not for the faint of heart. Even with the advantageous properties of the complementary models, estimators, and utility functions we had to know exactly what manipulations would lead to fruitful analytic results. Upon perturbing any of those inputs the convenience dissolves away and we typically have to consider numerical approximations.

Simulation, and Monte Carlo estimation of the expected utilities, offers a general strategy for approximating the calibration, but it can be extremely expensive in practice. A simulation for a single model configuration can be expensive on its own, and a frequentist calibration needs to simulate along a sufficiently fine grid of model configurations to quantify the full behavior of the expected utilities. This is complicated by the fact that we typically have to expand the grid all the way to the boundaries of the model configuration space where the worst case expected utility often hides. Still, simulation methods can be useful for low-dimensional observational models where relatively diffuse crude can be sufficient. As we increase the dimensionality of the model, however, we require exponentially finer grids, and exponentially increasing costs, that quickly overwhelm the available computational resources.

An unfortunate consequence of these challenges is that these systems with analytic results are often forced into applications where they have limited usefulness. The requisite observational model, for example, may be a poor approximation to the system being analyzed, or the requisite estimator may be irrelevant to the overall inferential goals of the analysis. In order to facilitate ignoring these complications, frequentist methods are often presented as black boxes for consuming data and automatically producing insights without any consideration of the implicit assumptions. The growing popularity of statistical software which can hide these assumptions in code often exacerbates the problem.

1.1 Probabilistic Modeling  
1.2 The True Data Generating Process  
1.3 The Observational Model

All of this said, there is one circumstance where relatively robust calibrations are available in closed form for a wide class of observational models. Under certain conditions, aggregating sufficiently-many independent observations results in model configurations that are reasonably well-approximated with Gaussian probability density functions. In this *asymptotic* regime we can then apply calculations similar to those introduced above to calibrate specific point estimators and set estimators. I discuss asymptotic results, and their limitations, more in Section Four.

2 Frequentist Inference  
3 Bayesian Inference  
4 Comparing Asymptotic Apples to Pears

## 3 Bayesian Inference

5 Conclusion  
6 Acknowledgements  
7 References  
8 License

Bayesian inference takes a more inclusive perspective on probability theory, allowing it to be used to model not only frequencies but also epistemological variation and weights over the model configuration space. Consequently we are free to use probability theory to quantify inferences within an observational model. Neighborhoods of model configurations that are more consistent with the available information would be allocated more probability, and those less consistent would be allocated less.

One immediate benefit of this approach is that it provides a much richer means of quantifying information about our model relative to the binary quantification of frequentist inference. A probability distribution encodes much more information than a step function!

Moreover, we don't have to choose a probability distribution over the model configuration space arbitrarily – there is a natural procedure for updating a *prior distribution* that encodes our domain expertise about the model into a *posterior distribution* that identifies those model configurations consistent with both our domain expertise *and* the observed data. Inferences, decisions, and predictions all follow from natural probabilistic operations.

### 3.1 Probabilistic Scholarship

A probabilistic quantification of what we know about our model would take the form of a probability distribution over the model configuration space that depends on any observed data.

## 1 Probabilistic Models

### 1.1 The Observational Model

### 1.2 The True Data Generating Process

### 1.3 The Observational Model

### 1.4 Model-Based Inference

## 2 Frequentist Inference

## 3 Bayesian Inference

## 4 Comparing Asymptotic and Preasymptotic Inference

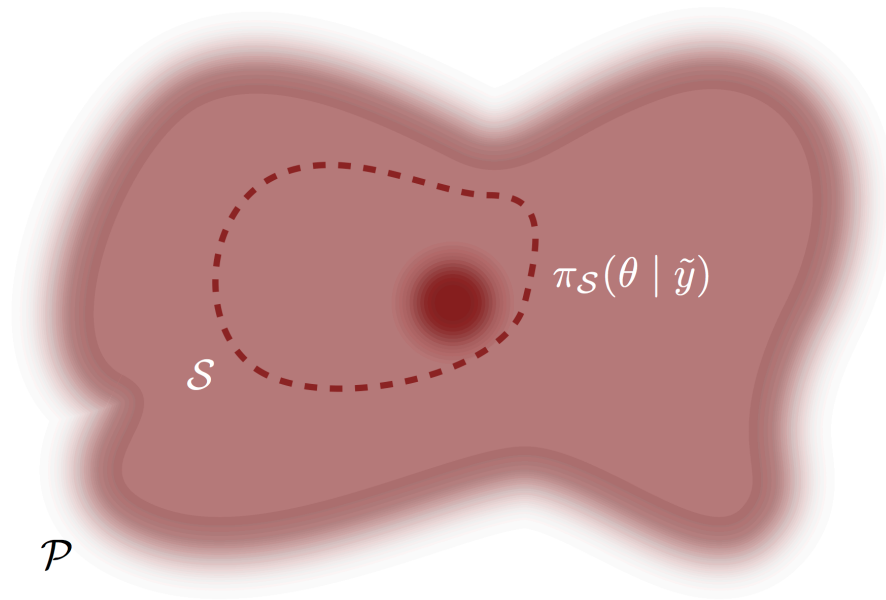
## 5 Conclusion

## Acknowledgements

## References

## License

## Original Computing Environment



Fortunately we don't have to construct such a distribution ourselves – it follows naturally from the rules of probability theory.

More specifically, once we allow probability distributions over the model configuration space then the observational model becomes more than just a collection of data generating processes – it becomes a *conditional probability distribution* over the product of the observational space and model configuration space,  $\mathcal{Y} \times \Theta$ , that we can specify with conditional probability density functions,

$$\pi_{\Theta}(\mathbf{y}; \theta) \rightarrow \pi_{\Theta}(\mathbf{y} \mid \theta).$$

Once we introduce a prior distribution we can apply *Bayes' Theorem* to invert this conditional relationship and give a posterior distribution that encodes our information about the system we're analyzing.

### 3.1.1 The Prior Distribution

A *prior model* is defined as a probability distribution over the model configuration space, typically with a probability density function  $\pi_{\Theta}(\theta)$ . The prior distribution encodes domain expertise about the model configurations in the observational model, and possibly even the context of the observational model relative to the true data generating process and latent observational process.

For example a prior distribution might assign less probability to model configurations that are extreme but not entirely impossible, suppressing their contribution to our resulting inferences. At the same time a prior distribution might suppress model configurations that, while not entirely impossible, we know stress the accuracy of our modeling approximations.

Importantly, prior distributions do not need to encode *all* of our domain expertise but rather just enough to ensure useful inferences. This can drastically simplify prior specification, for example by allowing priors to be relative ignorant about model configurations on which domain experts disagree. On the other hand, this means that in practice prior specification is best done in the context of the observational model. Consequently best practices for prior specification are most appropriately defined not in general but rather for specific modeling techniques where the complementary behavior of certain components of the observation model can be contrasted to certain component of the prior model.

## 3.1.2 The Likelihood Function

### 1 Probabilistic Modeling

If we specify the observational model with a family of probability density functions over the observational space,  $\pi_{\square}(y \mid \theta)$ , then the evaluation of those probability density functions at a specific observation,  $y$ , defines a *likelihood function*,

#### 1.2 The True Data Generating Process

#### 1.3 The Observational Model

The likelihood function maps each model configuration to a numerical quantification that increases for model configurations that are more consistent with the specific observation and decreases for those model configurations that are less consistent. In other words, the likelihood function quantifies the relative consistency of each model configuration with the observed data.

### 2 Frequentist Inference

Formally, the likelihood function provides only *relative* information – the absolute likelihood value depends on how we parameterize the observation space and hence doesn't have any well-posed meaning. Indeed from the Bayesian perspective it's not the likelihood function that quantifies consistency with an observation but rather the *likelihood ratio*.

### 5 Conclusion

Those not interested in the technical details should skip the rest of this section.

### Acknowledgements

Given a prior distribution and an observational model we can define a marginal probability distribution over the observation space,

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### Original Computing Environment

$$l_y: \Theta \rightarrow \mathbb{R}^+ \\ \theta \mapsto \pi_{\square}(y \mid \theta).$$

$$\pi_{\square}(y) = \int d\theta \pi_{\square}(y \mid \theta) \pi_{\square}(\theta).$$

The likelihood ratio is then defined as the Radon-Nikodym derivative of the distributions in the observational model with respect to this marginal probability distribution,

$$\square(y; \theta) = \frac{d\pi_{\square, y}}{d\pi_{\square}}(\theta) = \frac{\pi_{\square}(y \mid \theta)}{\pi_{\square}(y)}.$$

The likelihood ratio defines a numerical quantification of data consistency that, unlike the likelihood function, is independent of how we parameterize the observational space.

That said, the only difference between a likelihood function corresponding to a given parameterization of the observational space and the likelihood ratio is a constant that depends only on the observed data and not the model configurations. An arbitrary likelihood function and the likelihood ratio give the same relative quantifications of consistency. Because in typical practice we need only relative consistency to form well posed inferences, the difference between a likelihood function and the likelihood ratio is usually irrelevant.

## 3.1.3 The Posterior Distribution

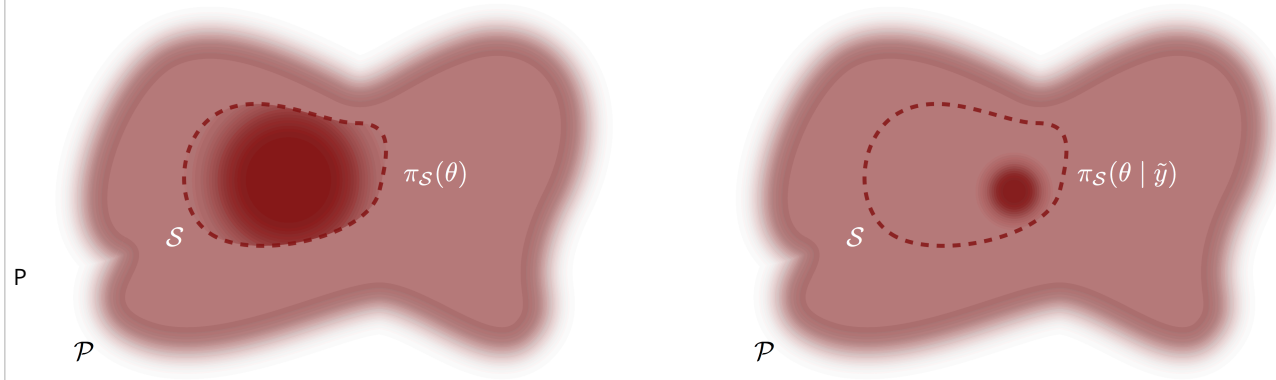
One we have a prior distribution and a likelihood function we can immediately construct a posterior distribution using Bayes' Theorem. Within a parameterization of the model configuration space, Bayes' Theorem defines the posterior density function as

$$\pi_{\square}(\theta \mid y) = \frac{\pi_{\square}(y \mid \theta)}{\int d\theta \pi_{\square}(y \mid \theta) \pi_{\square}(\theta)} \cdot \pi_{\square}(\theta) \\ \propto \pi_{\square}(y \mid \theta) \cdot \pi_{\square}(\theta).$$

The name Bayesian inference derives from this application of Bayes' Theorem, which can lead to confusion in practice because Bayes' Theorem a general result from probability theory and not unique to this application. In other words using Bayes' Theorem does not define Bayesian inference. Using probability distributions to quantify information about the model configuration space defines Bayesian inference.

Bayes' Theorem can be thought of as switching from one conditional probability density function, that specifying the observational model, to another, that specifying the posterior distribution. Bayes' Theorem can also be thought of a procedure for *updating* the prior distribution into a posterior distribution.

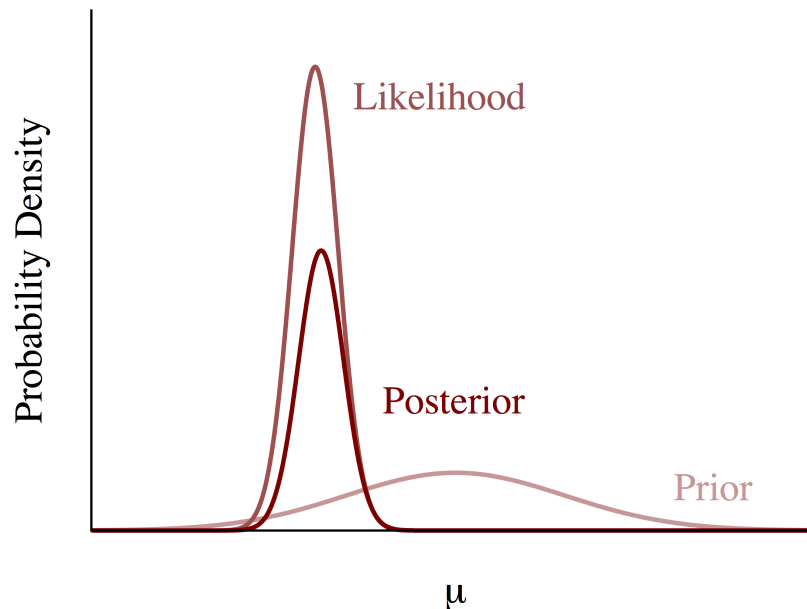
## 1.1 The Observational Process



## References

This latter perspective makes it more clear that this particular application of Bayes' Theorem implements a *learning* process. What we know after the observation, as encoded in the posterior distribution, is just the combination of what we knew before the observation, as encoded in the prior distribution, and what we learned from the observation, as encoded in the likelihood function. Bayes' Theorem ensures that all of this information is conserved and makes its way to the posterior distribution.

The posterior density function, and hence the posterior distribution itself, concentrates on those model configurations that are consistent with *both* our domain expertise, as encoded in the prior density function, and the observation, as encoded in the likelihood function. This manifests in three basic updating behaviors. *Contraction* occurs when the likelihood function is more informative than, but also consistent with, the prior distribution.



*Containment* occurs when the prior distribution is more informative than, but also consistent with, the likelihood function.

## 1 Probabilistic Modeling

1.1 The Observ

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5 Conclusion

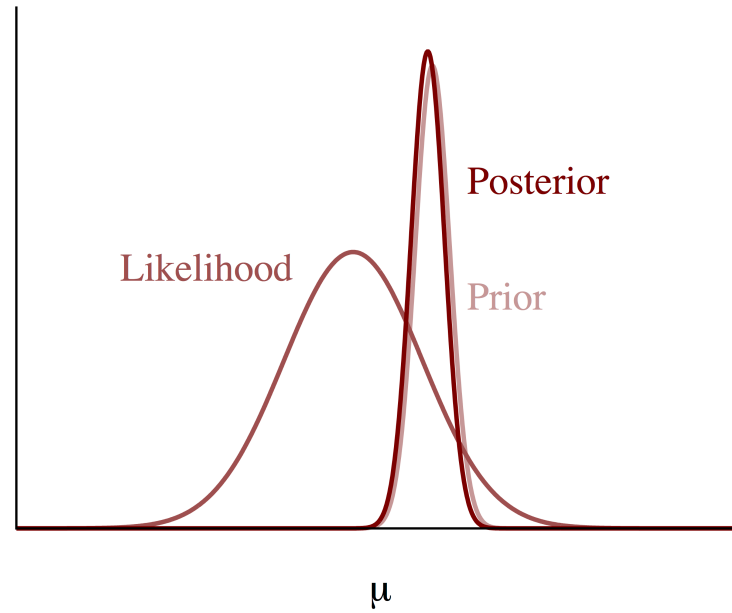
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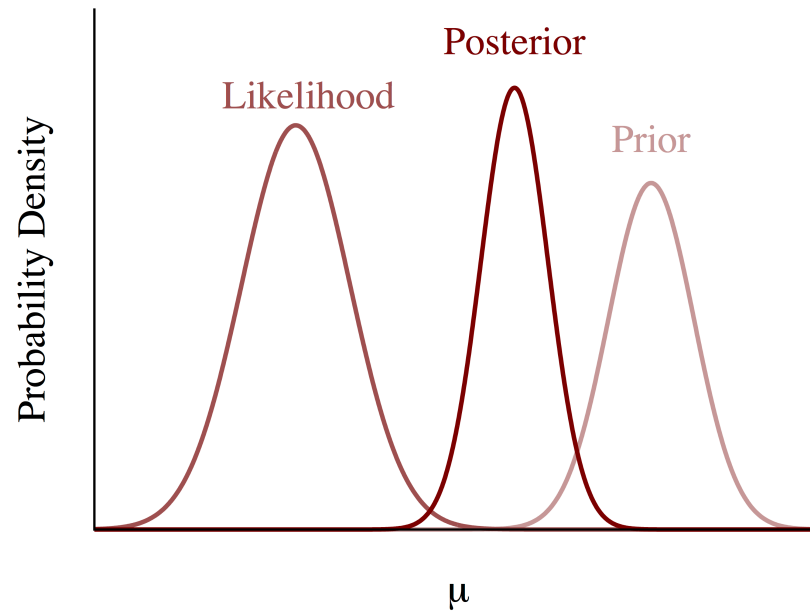
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Original Compu

Probability Density



*Compromise* occurs when there is tension between the information encoded in the likelihood function and the prior.



For example, consider the same observational model that we used to demonstrate frequentist inference,

## 1 Probabilistic Modeling

### 1.1 The Observational Process

### 1.2 The True Data Generating Process

Working through the algebra the joint probability density function simplifies to

### 1.3 The Observational Model

### 1.4 Model-Based Inferences

## 2 Frequentist Inference

where

## 3 Bayesian Inference

## 4 Comparing Asymptotic Apples to

Preasymptotic Oranges

## 5 Conclusion

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Moreover, let's presume that the scale parameter,  $\sigma$ , is known and utilize a prior distribution for the location parameter

Specified by a Gaussian probability density function,

$$\pi(y \mid \mu, \sigma) = \prod_{n=1}^N \square(y_n; \mu, \sigma).$$

$$\pi(y \mid \mu, \sigma) = (2\pi)^{-\frac{N}{2}} \sigma^{-N} \exp\left(-\frac{1}{2} \frac{N}{\sigma^2} (\langle y \rangle - \mu)^2\right) \exp\left(-\frac{1}{2} \frac{N}{\sigma^2} (\langle y^2 \rangle - \langle y \rangle^2)\right)$$

$$\langle y \rangle = \frac{1}{N} \sum_{n=1}^N y_n$$

$$\langle y^2 \rangle = \frac{1}{N} \sum_{n=1}^N (y_n)^2$$

$$\pi(\mu) = \square(\mu; m, s).$$

In this case the posterior distribution for the location parameter is specified with another Gaussian probability density function,

$$\pi(\mu \mid y) = \square\left(\mu; \frac{\langle y \rangle s^2 + m \frac{\sigma^2}{N}}{\frac{\sigma^2}{N} + s^2}, \sqrt{\frac{s^2 \cdot \frac{\sigma^2}{N}}{\frac{\sigma^2}{N} + s^2}}\right).$$

Depending on the observed data and the choice of  $m$  and  $s$ , the updating process can exhibit any of the three behaviors discussed above.

## 3.1.4 The Complete Bayesian Model

Given a particular prior distribution and observational model, Bayesian updating is a straightforward procedure.

Unfortunately the separation between the prior distribution and observational model is not always so clear in practice.

Consider, for example, a positivity constraint on one of the model configuration parameters corresponding to, say, the positivity of physical mass. Does this constraint belong in the observational model or should it be specified as a prior distribution that assigns zero probability to negative values? The resulting posterior distribution is the same either way.

Similarly, consider a model with unobserved latent structure between the model configuration parameters  $\theta$  and  $\phi$ ,  $\pi_{\square}(\theta \mid \phi)$ , such as that arising in hierarchical models and hidden Markov models. We could assume that the parameters  $\theta$  corresponding to hypothetically observable but missing data and define the observation model with the conditional probability density functions

$$\pi_{\square}(y, \theta \mid \phi) = \pi_{\square}(y \mid \theta) \pi_{\square}(\theta \mid \phi),$$

and a prior distribution with the probability density function

$$\pi_{\square}(\phi).$$

## 1 Probabilistic Modeling

At the same time, we could also define an observation model that consists of only the measured data,

### 1.1 The Observational Process

$$\pi_{\square}(y \mid \theta),$$

### 1.2 The True Data Generating Process

and then specify a prior distribution with the joint probability density function

### 1.3 The Observational Model

$$\pi_{\square}(\theta, \phi) = \pi_{\square}(\theta \mid \phi) \pi_{\square}(\phi).$$

### 1.4 Model-Based Inferences

Each specification is slightly different – the former presumes that the components of the model specified by the  $\theta$

parameters would vary from one observation to another, while the latter presumes that those components are static across

all observations. The corresponding posterior distributions for a *single* observation, however, are exactly the same.

## 2 Frequentist Inference

## 3 Bayesian Inference

## 4 Conclusion

## 5 Acknowledgements

## 6 References

## 7 License

## 8 Original Computing Environment

Consequently, as a Bayesian analysis it can be helpful to avoid discriminating between the observational model and the prior

model. Instead, focusing on the *complete Bayesian model*. More formally, before we observe any particular data the

prior distribution and observational model define a joint distribution over the observation space and the model

configuration space,  $Y \times \square$ , specified by the probability density function,

$$\pi_{\square}(y, \theta) = \pi_{\square}(y \mid \theta) \pi_{\square}(\theta).$$

$$\begin{aligned} \pi_{\square}(y, \theta) &= \pi_{\square}(y \mid \theta) \pi_{\square}(\theta) \\ &\propto \frac{\pi_{\square}(y \mid \theta)}{\pi_{\square}(y)} \pi_{\square}(\theta) \\ &\propto \pi_{\square}(\theta \mid y). \end{aligned}$$

Note that the constant of proportionality can also be written explicitly as an integral of the joint probability density,

$$\pi_{\square}(y) = \int d\theta \pi_{\square}(y \mid \theta) \pi_{\square}(\theta) = \int d\theta \pi_{\square}(y, \theta),$$

although in practice the unnormalized posterior almost always suffices.

One benefit of this joint perspective is that it facilitates model development. Once we separate the model configuration space into phenomenological, environmental, and probe parameters, for example, the full model can be decomposed into a sequence of conditional distributions that follows the generative structure of the true data generating process,

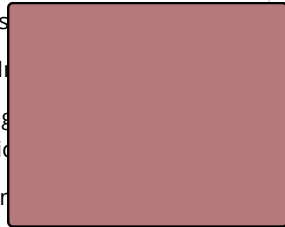
$$\begin{aligned} \pi_{\square}(y, \theta) &= \pi_{\square}(y, \theta_{\text{phenom}}, \theta_{\text{envir}}, \theta_{\text{probe}}) \\ &= \pi_{\square}(y \mid \theta_{\text{phenom}}, \theta_{\text{envir}}, \theta_{\text{probe}}) \\ &\quad \cdot \pi_{\square}(\theta_{\text{probe}} \mid \theta_{\text{phenom}}, \theta_{\text{envir}}) \\ &\quad \cdot \pi_{\square}(\theta_{\text{envir}} \mid \theta_{\text{phenom}}) \\ &\quad \cdot \pi_{\square}(\theta_{\text{phenom}}). \end{aligned}$$

Using this generative decomposition as scaffolding to construct the complete Bayesian model not only facilitates the development of coherent yet sophisticated models but also allows different parts of the model to be isolated and modeled separately. This can be hugely beneficial when the development is being done by a large team or collaboration and not any one individual.

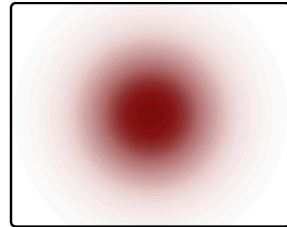


Another benefit of the complete Bayesian model is that it illuminates the *interaction* of the prior distribution and the likelihood functions for possible observed data. As discussed in Section 3.1.1 (sec:prior), this interaction is crucial for determining the amount of domain expertise that we need to encode in the prior distribution, and hence critical for principled model building.

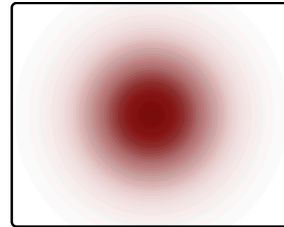
For example, if the likelihood functions derived from reasonable observations typically concentrate in a small neighborhood of the model configuration space then they will dominate the shape of the posterior density function and we get contractive behavior,



$\pi_S(\theta)$



$\pi_S(\tilde{y} \mid \theta)$

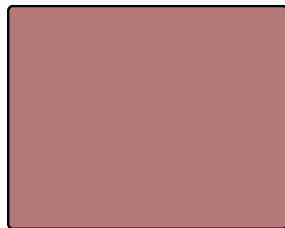


$\pi_S(\theta \mid \tilde{y})$

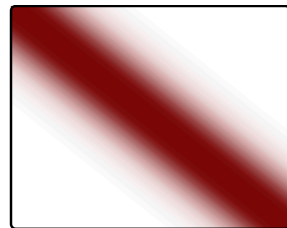
## References

In this case the information encoded in the posterior distribution is governed almost entirely by what we learn from the observation. Even vague priors are sufficient for useful inferences and we don't have to spend much time interrogating our original computing environment.

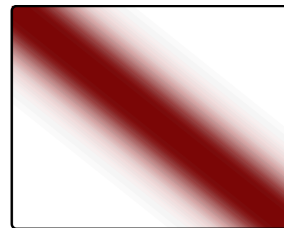
The possible likelihood functions, however, aren't always so well-behaved. In particular, our probe might not be sufficiently sensitive to the phenomena of interest at the scales approximated by the observational model. In that case the observational model is said to be *weakly-identified* and the likelihood functions tend to disperse over expansive regions of the model configuration space for most, if not all, observations. In circumstances like these the form of the posterior distribution is strongly influenced by the form of the prior distribution; a careless prior distribution will allow the weak identification of the observational model to propagate to the posterior distribution.



$\pi_S(\theta)$



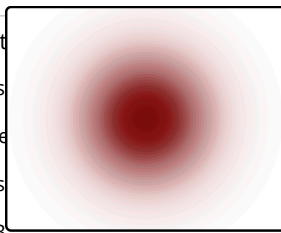
$\pi_S(\tilde{y} \mid \theta)$



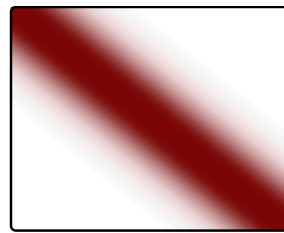
$\pi_S(\theta \mid \tilde{y})$

In order to compensate for the possibility of diffuse likelihood functions we need the prior distribution to incorporate at least a soupçon of domain expertise. At the very least we want the prior distribution to suppress extreme, although not entirely impossible, model configurations to ensure containment behavior in the Bayesian updating.

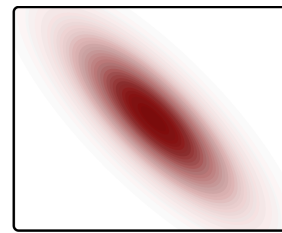
1 Probabilistic  
1.1 The Observed  
1.2 The True  
1.3 The Observed  
1.4 Model-Based



$\pi_S(\theta)$



$\pi_S(\tilde{y} | \theta)$



$\pi_S(\theta | \tilde{y})$

## 2 Frequentist Inference

### 3 Bayesian Inference

Specifying prior distributions that are robust to possible pathologies in the likelihood functions derived from the observational model is a key to robust analyses. This specification is itself greatly facilitated by not modeling the prior distribution of the observational model independently but rather together through the development of a complete Bayesian model.

## 5 Conclusion

### Acknowledgements

## 3.2 Employing Our Education

### References

Bayes' Theorem ensures that any information learned by our model is encoded in the posterior distribution. Consequently any inferential query we might pose must be answered with only valid manipulations of the posterior distribution. In probability theory, however, all valid manipulations take the form of the evaluation of expectation values or the construction of pushforward distributions. This seemingly limited set of inferential outputs provides a rich diversity of applications.

### 3.2.1 Making Inferences

Most inferences take the form of posterior *expectation values* of certain functions. In other words, the question we ask is encoded in a real-valued function over the model configuration space,  $g : \Theta \rightarrow \mathbb{R}$ , and the corresponding answer is given by the corresponding posterior expectation value,

$$\mathbb{E}[g(\theta)] = \int d\theta \pi_{\square}(\theta | y) g(\theta).$$

For example we might ask *where* in the model configuration space our posterior concentrates. We can answer this with posterior means or even posterior medians, which are derived from the difference of expectation values. Similarly the *breadth* of this concentration can be quantified by posterior standard deviations or perhaps posterior tail quantiles. More sophisticated questions are answered with expectation values of more sophisticated functions.

The key step in *implementing* a Bayesian analysis is *evaluating* the posterior expectation values for whatever functions are deemed relevant to the analysis.

If the posterior density can be worked out analytically then certain expectation values are given exactly. The model introduced in Section 3.1.3, for example, yields the Gaussian posterior density function,

$$\pi(\mu | y) = \square \left( \mu; \frac{\langle y \rangle s^2 + m \frac{\sigma^2}{N}}{\frac{\sigma^2}{N} + s^2}, \sqrt{\frac{s^2 \cdot \frac{\sigma^2}{N}}{\frac{\sigma^2}{N} + s^2}} \right).$$

The posterior mean can then be immediately identified as

## 1 Probabilistic Modeling

### 1.1 The Observational Process

while the posterior standard deviation is given by

### 1.2 The True Data Generating Process

### 1.3 The Observational Model

### 1.4 Model-Based Inferences

## 2 Frequentist Inference

Similarly, quantiles can be readily computed using the inverse Gaussian cumulative distribution function, which is available in most computational statistical libraries.

## 3 Bayesian Inference

Even with a closed-form posterior density function, however, the expectation value of more sophisticated functions will not always be available analytically. Moreover, closed-form posterior density function rely on very simple models and hence are unrealistic in all but the simplest applications. In practice we instead rely on probabilistic computational methods to

## 4 Comparing Asymptotic

approximate various posterior expectation values. Markov chain Monte Carlo, for example, is a paramount technique which

can consume the joint density function that specifies the complete Bayesian model and return estimates of posterior expectations along with quantifications of the estimator errors.

## References

The term "fit" is often used to colloquially denote statistical computation, derived from a time when statistical computation

referred to finding a single model configuration that best fit the observed data. These days statistical computation is much

more complex and careless use of the term promotes confusion. Obtaining a normalized posterior density function, for example, *cannot* be the entirety of a Bayesian fit because it is useful for evaluating at best only some expectation values. A proper Bayesian fit is instead the computation, or more realistically estimation, of *all* expectation values relevant to a given application. Equivalently one might define a Bayesian fit as the generation of intermediate objects, such as posterior samples, that can be used after the fact to estimate any expectation value.

## 3.2.2 Making Decisions

The posterior distribution also provides a formal means for the robust decision making that originally motivated our foray into statistical inference.

As we saw in Section 1.3.3 a model-based utility function

$$U : A \times \Theta \rightarrow \mathbb{R}^+ \\ (a, \theta) \mapsto U(a, \theta).$$

isn't immediately applicable for informing decisions because of the dependence on the unknown model configuration. In a Bayesian analysis, however, the posterior distribution quantifies the relevance of each possible model configuration. Consequently we can compute *expected utilities* as posterior expectation values,

$$\overline{U}(a, y) = \mathbb{E}_{\pi_{\square}(\theta|y)}[U(a, \theta)] = \int d\theta \pi_{\square}(\theta|y) U(a, \theta).$$

Note the implicit dependence on the observation due to its influence on the posterior distribution.

The optimal decision *in the context of the available information* is then given by taking the action that maximizes these expected utilities. This defines the *Bayesian decision making process*

$$a^*(y) = \operatorname{argmax}_{a \in A} U(a, y).$$

Mathematically Bayesian decision theory is straightforward and a natural consequence of using probability theory to quantify information. At the same time it can be a conceptual paradigm shift for many because of its explicit consideration

of the *uncertainty* inherent in our inferences. Indeed many common fallacies arise from ignoring inferential uncertainty, or presuming models that poorly approximate the true data generating process. Going through the steps of developing a Probabilistic Modeling Bayesian model and computing expected utilities is a powerful way of avoiding common mistakes and facilitating more robust decisions.

For example, let's say that our goal is to report a point summary of the posterior distribution,  $\hat{\theta}$ , and we want that summary to minimize the squared difference to the best model configuration. In that case the utility function would take the form

$$U(\hat{\theta}, \theta) = -(\hat{\theta} - \theta)^2.$$

From this, the expected utility would become

$$\begin{aligned}\overline{U}(\hat{\theta}, \tilde{y}) &= \int d\theta \pi_{\square}(\theta \mid \tilde{y}) U(\hat{\theta}, \theta) \\ &= - \int d\theta \pi_{\square}(\theta \mid \tilde{y}) (\hat{\theta} - \theta)^2 \\ &= -(\hat{\theta} - \mathfrak{m}(\tilde{y}))^2 - \mathbb{V} \tilde{y},\end{aligned}$$

where  $\mathfrak{m}(\tilde{y})$  is the posterior mean and  $\mathbb{V}(\tilde{y})$  is the posterior standard deviation. This expected utility is maximized when we report the posterior mean, which defines the optimal point summary,

$$\hat{\theta}^* = \mathfrak{m}(\tilde{y}).$$

A more elaborate circumstance might consider how to optimally dose a new medication. Strong doses might be more effective, but they also suffer from increased, possibly even lethal, side effects. Weaker doses are safer but may not be sufficiently effective at treating a disease. The overall utility of a given dosing will then depend on how the medication interacts with the physiology of a patient, or typically the expected population of patients that might receive this treatment. With a sufficiently sophisticated model we can infer these interactions, compute the expected utility of each action, and then make our decision about how best to dose the treatment.

Bayesian decision theory is a powerful framework but it strives only to make an optimal decision in the context of the available information. There is no guarantee that the resulting decision-making process will perform well across the range of observations distributed according to the true data generating process. For that we need to go one step further and *calibrate* our decision-making process over some incidence of reasonable observations.

## 3.2.3 Making Predictions

If the latent observational process is stationary then predictions for future observations are no different from post hoc predictions of the observed data. In particular, predictions are straightforward to make given a particular model configuration; the predictive distribution is simply the corresponding data generating process.

In Bayesian inference we could isolate a single model configuration with a carefully designed expectation value, such as the posterior mean for each parameter. That isolation, however, would prevent us from utilizing all of the information available to our model. Instead we can construct predictions sensitive to *all* of the possible model configurations by convolving a probability distribution over the model configuration space with each of the possible data generating processes. Those configurations allocated more probability will then more strongly influence the resulting predictive distribution relative to those allocated less probability.

Because we have two natural probability distributions over the model configuration space available in Bayesian inference, the prior distribution and posterior distribution, we have two natural predictive distributions.

### 3.2.3.1 The Prior Predictive Distribution

One possibility is to average the data generating processes in the observational model with respect to the prior distribution.

This convolution yields the *prior predictive distribution* which is specified by the probability density function

## 1 Probabilistic Modeling

### 1.1 The Observational Process

### 1.2 The True Data Generating Process

Note that the integrand is the probability density function that specifies the complete Bayesian model. This implies that we can equivalently derive this predictive distribution as a marginal of the complete Bayesian joint distribution,

### 1.3 The Observational Model

### 1.4 Model-Based Inferences

## 2 Frequentist Inference

## 3 Bayesian Inference

The prior predictive distribution is most strongly influenced by those model configurations most consistent with the domain expertise encoded in the prior distribution. In other words the prior predictive distribution quantifies our best understanding of the scope of reasonable observations using only the domain expertise that we've introduced. This allows

us to make predictions *before* we've made any observations, which is useful when we want to study how our inferences and

subsequent decision making processes might play out before we probe the latent system.

## 5 Conclusion

## Acknowledgements

In practice we typically cannot construct the prior predictive density function in closed form, but we have a few ways of

generating samples from the predictive distribution. One approach is to employ ancestral sampling: first we exactly sample

a model configuration from the prior distribution,

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## Original Computing Environment

$$\pi_S(y) = \int d\theta \pi_{\square}(\theta) \pi_{\square}(y | \theta).$$

$$\pi_S(y) = \int d\theta \pi_{\square}(y, \theta).$$

$$\theta \sim \pi_{\square}(\theta),$$

and then we simulate an observation from the corresponding data generating process,

$$y \sim \pi_{\square}(y | \theta).$$

If we cannot exactly sample from both the prior distribution and the data generating processes in the observational model, or if we specified only the complete Bayesian model and don't have distinct prior and observational models, then we cannot implement ancestral sampling. In that case we can run Markov chain Monte Carlo on the joint distribution to generate correlated samples from the joint distribution and then project out the sampled model configurations,

$$\{\theta, y\} \rightarrow \{y\}.$$

Once we have prior predictive samples we can estimate prior predictive expectation values, pushforward distributions, or even further convolutions.

### 3.2.3.2 The Posterior Predictive Distribution

The other possibility is to average the data generating processes in the observational model with respect to the posterior distribution. This convolution yields the *posterior predictive distribution* which is specified by the probability density function

$$\pi_S(y | y) = \int d\theta \pi_{\square}(\theta | y) \pi_{\square}(y | \theta).$$

Because the posterior predictive distribution uses the observational model twice, once to inform the posterior and again for the predictions, it requires an explicit decomposition of the complete Bayesian model into a prior model and an observational model.

The posterior predictive distribution is influenced by the model configurations consistent with not just the relevant domain expertise but also the particular data we've observed. This makes it ideal for informing predictions about future interactions with the latent system.

Comparisons between the posterior predictive distribution and the observed data also provide a measure of how well our model approximates the latent system. Although we can't explicitly form these comparisons in practice, *estimates* of these comparisons form the basis of all predictive model validation procedures, including information criteria and cross validation. **1.3 The Observational Model**

As with the prior predictive density function, the posterior predictive density function is rarely available in closed form. If we have already generated posterior samples, say through a Markov chain Monte Carlo fit, then we can generate posterior predictive samples,  $y^*$ , through ancestral sampling,

1.4 Model-Based Inferences

2 Frequentist Inference

3 Bayesian Inference

## 3.3 Bayesian Calibration

4 Comparing Systems and Applications

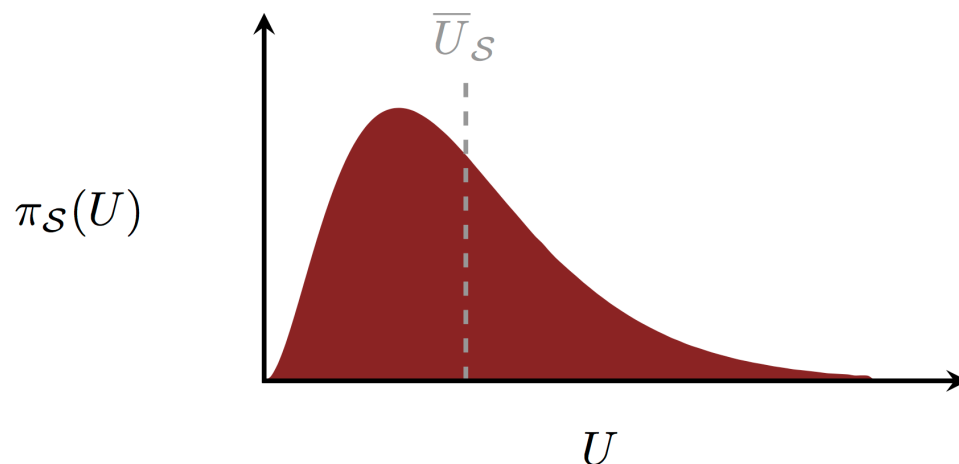
Preasymptotic Oranges

Ideally the true data generating process, or at least the best approximation to it within our observational model, would fall into the neighborhood of high posterior probability. That accuracy would ensure that whatever conclusions we draw from our model, in particular the decisions informed by those inferences, would be faithful to the real latent system being analyzed.

In general, however, there are no such guarantee in Bayesian inference. The posterior distribution naively follows the information encoded in the model and the observation. Without a carefully designed model the posterior distribution is vulnerable to being led astray by irrelevant quirks in a given observation, ending up far away from the desired model configurations. Bayesian inference provides a procedure for constructing inferences but it offers no general guarantees on the overall behavior of those inferences. If we want to ensure robust inferences then we at least need to attempt some *calibration* in order to determine how we expect our model to perform.

Following Section 1.3.3 we might consider inferential outcomes of our model that take the form of a decision-making process,  $a(y)$ . For example, we could consider  $a(y)$  as a particular property of the posterior distribution, or the outcome of Bayesian decision making process. In that case a utility function  $U(a(y), \theta)$  provides a calibration for a specific observation and model configuration. We can then use the data generating process corresponding to a given model configuration to extend that calibration across possible observations, but so far we've had limited ways of extending the calibration across the unspecified model configuration. For example, in the frequentist setting we were limited to calibrations based on worst case expected utility.

In a Bayesian analysis we can use the joint distribution corresponding to the complete Bayesian model to account for both the unknown observation *and* the unknown model configuration. We can push the joint distribution through the utility function to construct an entire probability distribution of potential outcomes,  $\pi_{\mathcal{S}}(U)$ ,



and then define calibrations such as expected utilities,  
1 Probabilistic Modeling

1.1 The Observational Process

1.2 The True Data Generating Process

In practice we can implement this calibration by sampling from the joint distribution and then evaluating the utility function,  
1.3 The Observational Model

1.4 Model-Based Inferences

2 Frequentist Inference

3 Bayesian Inference

Put in words, we can calibrate a Bayesian analysis by repeatedly simulating model configurations from the prior distribution, observations from the corresponding data generating process, running our analysis to completion, and then scrutinizing the resulting likelihood functions, posterior distributions, and decision outcomes in the context of the simulated ground truth.  
4 Comparing Asymptotic Apples to Presymptotic Oranges

5 Conclusion

The conclusions we draw from these calibrations will always be limited by the quality of the complete Bayesian model. If we are not confident that our observational model reasonably approximates the true data generating process, or that our prior model encodes sufficient domain expertise to regularize any misbehaving likelihood functions, then of what use are these calibrations? In practice these calibrations serve as a way of *bootstrapping* model development. Starting with an initial crude model we can use these calibrations to identify the limitations of that model and inform improvements such as more sophisticated observational models, more precise domain expertise, or even more robust methods for computing expectations. This information is critical to a *principled Bayesian workflow* for model development  
Acknowledgments

References

Models

Original Computing Environment

([https://betanalpha.github.io/assets/case\\_studies/principled\\_bayesian\\_workflow.html](https://betanalpha.github.io/assets/case_studies/principled_bayesian_workflow.html)).

At the same time, the complete Bayesian model used for constructing the calibration simulations doesn't have to be the same model used to construct our inferences. We can, for example, use a more sophisticated, and possibly more expensive, model to calibrate a more approximate model that we aim to employ in a given application. Indeed this use of multiple models is not uncommon in scientific practice.

Bayesian calibration is surprisingly general, and it spans a wide range of useful analyses. Here I introduce just a few of the techniques that fall under this general approach.

### 3.3.1 Calibrating Predictions

One immediate application is the study of the simulated data itself, which we can equivalently think of as samples from the prior predictive distribution. Tension between the behavior of this simulated data and our expectations indicates that our prior model needs to integrate more domain expertise.

This *prior predictive check* has a long history in Bayesian inference, going all the way back to Good's "device of imaginary results" [12]. A natural extension is to study not just the simulated data but also the behavior of the resulting likelihood functions, inspecting for poor identification that might require more domain expertise.

### 3.3.2 Calibration Posterior Behaviors

Studying the properties of potential posterior distributions resulting from these calibration simulations allows us to identify common pathologies, such as overfitting and poor identifiability, that limit the utility of any resulting inferences. In addition to evaluating specific decision making processes we can also study the more general properties that can identify suspicious behavior.

$$\overline{U}_{\square} = \int dy d\theta \pi_{\square}(y, \theta) U(a(y), \theta).$$

$$\begin{aligned}\theta &\sim \pi_{\square}(\theta) \\ y &\sim \pi_{\square}(y \mid \theta) \\ \tilde{U} &= U(a(y), \theta).\end{aligned}$$

For example, if  $\theta$  is the true model configuration then we can quantify how closely the posterior distribution is to the true

model configuration with the z-score for each parameter,

1 Probabilistic Modeling

1.1 The Observational Process

1.2 The True Data Generating Process

1.3 The Observational Model

where  $m_{\text{post}}$  is the marginal posterior mean and  $V_{\text{post}}$  is the marginal posterior variance. Smaller z-scores indicate that the posterior distribution concentrates around the true model configuration while larger values indicate concentration away from the true model configuration.

1.4 Model-Based Inferences

2 Frequentist Inference

At the same time the *contraction* of a given parameter,

3 Bayesian Inference

4 Comparing Asymptotic Apples to

Preasymptotic Oranges

5 Conclusion

provides one possible quantification for how much the posterior is influenced by the information encoded in the observed likelihood function. Contraction near zero indicates that the simulated data provided little information beyond the domain expertise encoded in the prior distribution, while contraction near one indicates a likelihood function that strongly informs that parameter.

Acknowledgements

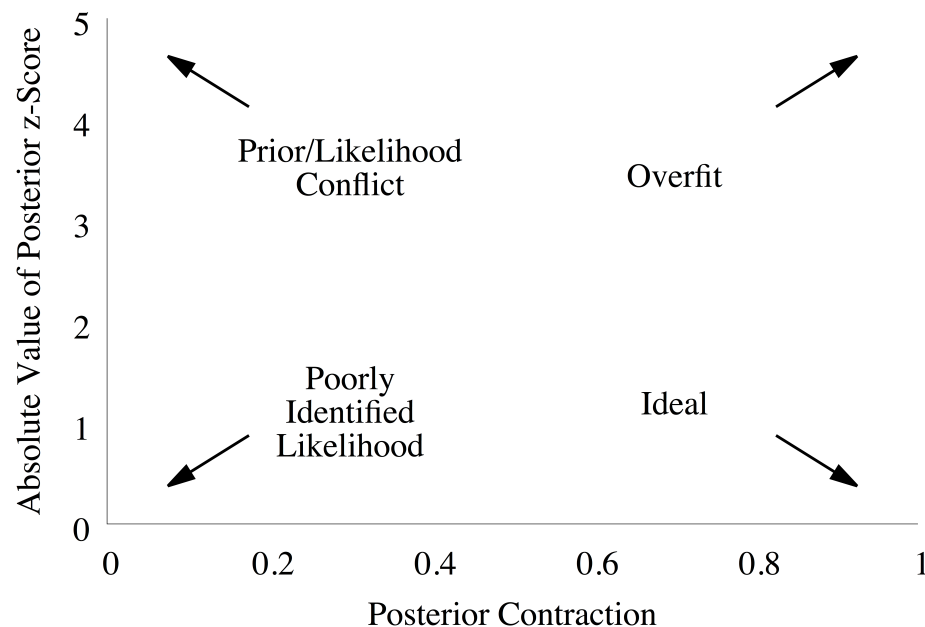
References

Convergence

Conveniently, these two posterior expectations are sensitive to an array of pathological behavior that can compromise our inferences. The ideal behavior concentrates towards strong contraction and small z-scores, indicating a posterior

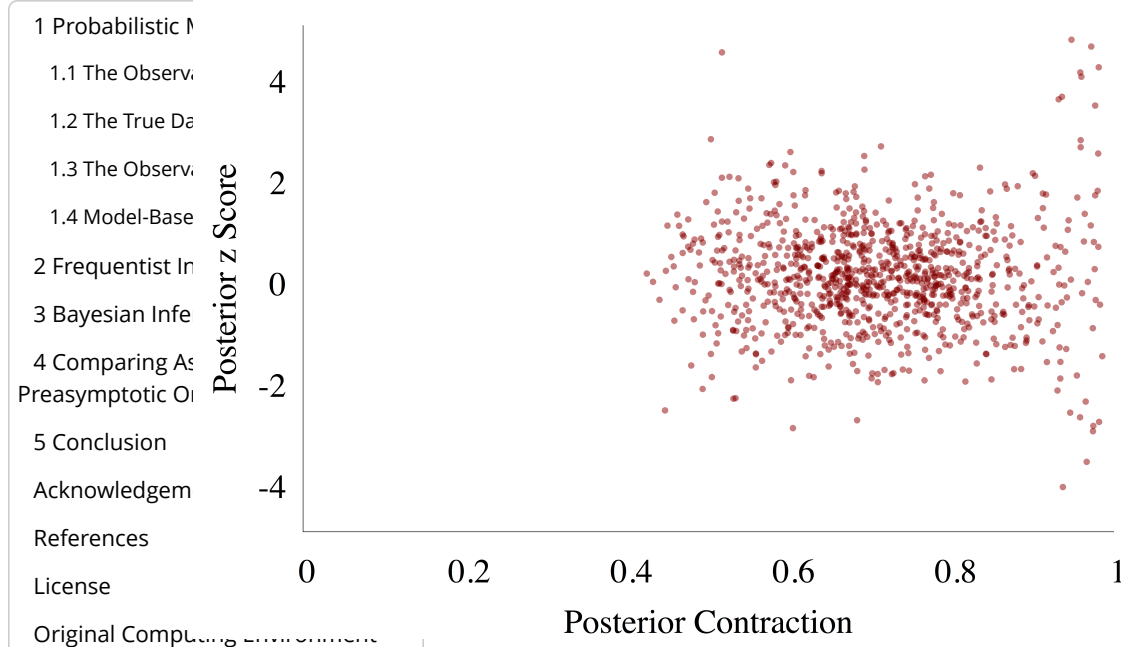
Original Computing Environment

distribution that strongly concentrates around the true model configuration. A posterior distribution with strong contraction and a large z-score, however, indicates a model that overfits to irrelevant structure of the simulated data and is unable to accurately recover the true model configuration. Similarly a posterior distribution with weak contraction and a small z-score indicates a model that is poorly informed by the simulated data. Finally a posterior distribution with weak contraction and a large z-score indicates a fit where the likelihood function and prior distribution disagree with each other.



Consider, for example, this output from simulations using a simple regression model. Each point records the z-score and contraction of the posterior distribution derived from one of the simulations from the complete Bayesian model.





We see that most of the resulting posterior distributions are reasonably well-behaved with small z-scores and moderate contraction. Occasionally, however, the posterior distribution chases some irrelevant structure in the data resulting in strong contraction and much larger z-scores. This indicates that are model isn't quite as robust as we might hope, and it has the nasty habit of being its most confident exactly when it's the most wrong.

### 3.3.3 Calibrating Posterior Computation

We can even use Bayesian calibration to quantify how faithfully our computational tools represent the models that they're fitting.

As we fit each posterior distribution for each simulated observation we can evaluate the computational performance. At a very high level we can see if it even successfully ran to completion, if it never finished within a reasonable time frame, or if it encountered a software failure due to bugs, exhausted memory resources, and the like. Going further we can evaluate any self-diagnostics in completed fits, such as the potential scare reduction factor for general Markov chain Monte Carlo algorithms or divergences and energy fractions of missing information in Hamiltonian Monte Carlo. If our tool proves robust to all of the potential posterior distribution behaviors across the simulated observations then we have some assurance that it should perform well enough for any real observations, at least if our model is sufficiently accurate.

What can we do, however, if our computational tool isn't self-diagnostic? Fortunately Bayesian inference provides a subtle consistency property that allows us to check *any* method capable of generating posterior samples. Regardless of our particular model, averaging the posterior distributions fit from observations simulated from the prior predictive distribution will *always* recover the prior distribution,

$$\pi_{\square}(\theta') = \int dy d\theta \pi_{\square}(\theta' | y) \pi_{\square}(y, \theta).$$

## 1 Probabilistic Modeling

Particular case this implies that the ensemble of posterior samples,  $\theta^*$ ,

### 1.2 The True Data Generating Process

$$\tilde{\theta} \sim \pi_{\square}(\theta)$$

### 1.3 The Observational Model

$$y \sim \pi_{\square}(y | \tilde{\theta})$$

### 1.4 Model-Based Inferences

$$\theta^* \sim \pi(\theta | y),$$

With low the inference distribution as as the incident samples from the prior distribution,  $\tilde{\theta}$ . Any deviation between the two samples indicates that either our simulations are incorrect, our fit is using an inconsistent model from the simulations, or that our computational tool is generating biased samples.

### 4 Comparing Asymptotic Apples to

*Simulated-based calibration* [13] compares the ensemble posterior sample and the prior sample using *ranks*. For each simulated observation we generate  $R$  samples from the corresponding posterior distribution,

### 5 Conclusion

### Acknowledgements

### References

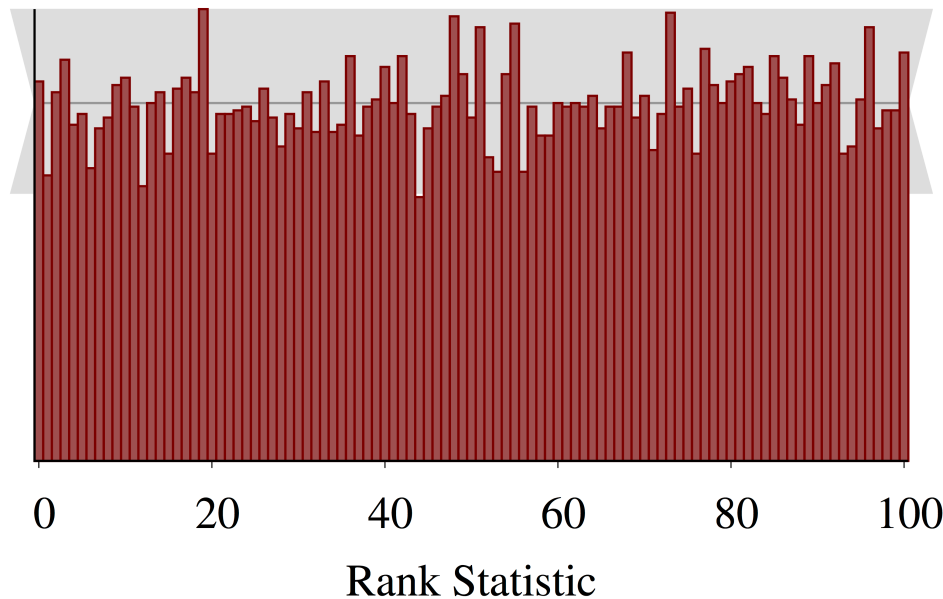
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and compute the rank of the prior sample within the posterior samples, i.e. the number of posterior samples larger than the prior sample,

$$\begin{aligned} \tilde{\theta} &\sim \pi_{\square}(\theta) \\ y &\sim \pi_{\square}(y | \tilde{\theta}) \\ (\theta_1^*, \dots, \theta_R^*) &\sim \pi(\theta | y), \end{aligned}$$

$$\rho = \# \left\{ \tilde{\theta} < \theta_r^* \right\}.$$

If the ensemble posterior sample and the prior sample are equivalent then these ranks will be uniformly distributed. The uniformity of the ranks itself is straightforward to analyze visually by histogramming the ranks for each parameter in the model configuration space.



Here the grey band demonstrates the expected variation of ranks distributed according to an exact uniform distribution.

Deviation outside of these bands, especially systematic deviation of many bins at once, indicates computational problems.

## 1 Probabilistic Modeling

See [13] for further discussion of common systematic deviations and their interpretations.

### 1.1 The Observational Process

### 1.2 The True Data Generating Process

### 1.3 The Observational Model

### 1.4 Model-Based Inferences

So far our discussion of frequentist and Bayesian inference has neglected a ubiquitous, and often abused, concept that

completely changes the discussion of how we implement inference in practice: *asymptotics*. In this final section let's use

what we've learned so far as a foundation on which we can properly introduce asymptotics and discuss its practical

limitations.

## 4 Comparing Asymptotic Apples to Preasymptotic

Consider an observation comprised of many identical and independently generated components,  $y = \{y_1, \dots, y_N\}$  with

the total observation space defined as the product of each component observation space,  $\times_{n=1}^N Y$ .

## 5 Conclusion

As the number of these components,  $N$ , grows our observation should assimilate more information. Eventually only those

model configurations that closely approximate the true data generating process should have any chance of being inferred

as consistent with the observation. In other words, the likelihood function should concentrate around the true data

generating process, at least if it's contained within the observational model.

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At the same time, when we consider an observation with an increasing number of components the idiosyncratic properties

of any particular component observation should average out. How the likelihood function concentrates around the true

data generating process should largely be irrelevant of the particular details of any given observation.

These intuitions are formalized in the theory of *asymptotics*. Here we assume that the likelihood function for any

observation is smooth and then consider the *maximum likelihood estimator*,

$$\theta_{\text{ML}}(y) = \underset{\theta \in \Theta}{\operatorname{argmax}} \pi_{\square}(y; \theta).$$

If the observational model contains the true data generating process,  $\pi^{\dagger}(y) = \pi(y; \theta^{\dagger})$  for some true parameters  $\theta^{\dagger}$ , then

under some technical conditions the distribution of maximum likelihood estimates converges towards  $\theta^{\dagger}$  as the number of

component observations increases. In particular, for sufficiently large  $N$  the maximum likelihood estimator approximately

follows a distribution specified by a Gaussian probability density function,

$$J(\theta_{\text{ML}}(y)) \cdot (\theta_{\text{ML}}(y) - \theta^*) \sim \square(0, 1),$$

where  $J$  is the *Fisher information matrix* which evaluates the expectation value of the Hessian of the log likelihood function,

$$J_{ij}(\theta) = \int dy \pi_{\square}(y; \theta) \frac{\partial^2}{\partial \theta_i \partial \theta_j} \log \pi_{\square}(y; \theta).$$

A common temptation is to write this result as

$$\theta_{\text{ML}}(y) \sim \square(\theta^*, J(\theta_{\text{ML}}(y))),$$

but this is technically incorrect as the pushforward distribution for the estimator cannot depend on the estimator itself! The

slightly awkward form used above gathers the maximum likelihood estimator dependencies together to avoid this problem.

Once this asymptotic behavior takes hold, any expected utility that depends only on the maximum likelihood estimator or

the Fisher information becomes a Gaussian integral which will be straightforward to approximate, if not outright derive in

closed-form, for many utility functions. Moreover, because the distribution for the maximum likelihood estimator depends

on only the true model configuration, we don't have to worry about figuring out a worst case expected utility. All of the

expected utilities will be the same!

consequently in this asymptotic limit we can construct point estimators and confidence sets with known calibrations regardless of the specifics of the observational model. To implement them in practice all we have to do is compute a straightforward maximization of the likelihood function. If we're being good we'd also verify that the observation has enough independent components for the asymptotic limit to be a reasonable approximation for *all* possible true model configurations.

1.2 The True Data Generating Process

Under mild conditions this behavior also carries over to Bayesian inference. Provided that the prior is smooth and doesn't assign zero probability to any neighborhood of the model configuration space then the *Bernstein-von Mises Theorem* shows that the posterior distribution manifests the same concentration around the true data generating process that the maximum likelihood estimator does. In this case most posterior expectation values become much easier to approximate, if not evaluate exactly.

2 Frequentist Inference

3 Bayesian Inference

This asymptotic limit is a paradise relative to the fragile world that I've carefully introduced above. The problem is that verifying for what  $N$  it gives an approximation is no trivial task. Simpler models will converge to the asymptotic limit faster and hence be well approximated with smaller  $N$ , while more complex models will typically require larger  $N$ . Working out the exact details of this convergence for each model configuration in the observational model, however, requires serious mathematical and computational tools.

5 Conclusion

Acknowledgements

Perhaps more importantly, the assumption that the observational model contains the true data generating process can restrict the applicability of the asymptotic limit altogether. Recall that observational models are themselves often crude approximations to the true data generating process. For small  $N$  the observation will be only weakly informative and we won't be able to resolve the system well enough to notice the approximation. As  $N$  increases, however, our observations become more informative enough that we can resolve the limitations of the observational model. In other words, as we try to escape to the asymptotic limit the assumption that the observational model contains the true data generating process become harder and harder to justify.

Consider, for example, an observational model where we ignore any heterogeneity in the processes that generate each component observation. In this case our model might need only a few parameters homogenous to all the components. For sufficiently many components in the observation the asymptotic limit will well approximate this homogenous observational model. In that limit, however, our inferences for the homogenous parameters become precise enough that we start to resolve the heterogeneity that we ignored. Even though the homogenous observational model converges to the asymptotic limit, it *diverges* from the true data generating process and the limit becomes largely irrelevant.

When the observational model is an approximation, the ultimate utility of the asymptotic limit depends on the existence of an intermediate  $N$  for which the observational model converges close enough to the asymptotic limit without diverging too far from the true data generating process. Demonstrating that such an  $N$  exists in a particular application, and then identifying it if it does, is an extremely challenging task.

Still, frequentist asymptotics have become the norm in many applied fields because of the ease of their implementation, along with the ease with which the underlying assumptions can be ignored. Indeed frequentist inference has become so tightly bound to the asymptotic limit in much of the academic literature it becomes the basis for unfounded comparisons to Bayesian methods.

A common refrain is that frequentist methods are much less computationally demanding than Bayesian methods. This comparison, however, is not between the general application of the two methodologies but rather between the application of frequentist methods in the asymptotic limit and the general application of Bayesian methods *without* the luxury of that limit. A single optimization through the model configuration space is undoubtable less computationally demanding than the survey of the model configuration space needed to accurately approximate posterior expectation values, but this isn't a proper comparison.

The asymptotic limit drastically simplifies the structure of the observational model, reducing the burden of *both* frequentist and Bayesian computation. In the asymptotic limit we no longer need to rely on general Bayesian tools for computing posterior expectations and can instead utilize approximate algorithms that exploit the asymptotic structure. If the limit well approximates the observational model in a given application then any fair comparison has to allow both methodologies to take advantage of it. In that case the computational burden of both approaches becomes nearly equivalent.

On the other hand, if the asymptotic limit poorly approximates the observational model then we have to rely the general frequentist and Bayesian methodologies. Here we can no longer rely on the asymptotic calibration of maximum likelihood estimators and instead have to engineer a bespoke estimator and go through the costly calibration process in its entirety. In comparison, building and calibrating a Bayesian model is often much less computationally demanding.

## 5 Conclusion

Bayesian inference is a versatile methodology for fusing information from observations and domain expertise into coherent inferences and decision-making processes. The probabilistic foundations of the methodology ensure that the mechanisms for evaluating, analyzing, and criticizing inferences are all implemented by natural operations in probability theory.

Conceptually this makes Bayesian computation particularly well-suited to automation once a complete Bayesian model has been specified. In practice automated computation can't quite be realized, but with the careful utilization of modern tools we can get impressively close. At the same time, carefully designed probabilistic programming languages provide the means for practitioners to specify the broad extent of models that arise in applications.

Together probabilistic programming and state-of-the-art Bayesian computational methods provide a platform for implementing Bayesian inference in practice. In particular, their combination is the basis for Stan [14].

## References

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## References

- [1] CASELLA, G. and BERGER, R. L. (2002). *Statistical inference*. Duxbury Thomson Learning.
- [2] LEHMANN, E. L. and CASELLA, G. (2006). *Theory of point estimation*. Springer Science & Business Media.
- [3] KEENER, R. W. (2011). *Theoretical statistics: Topics for a core course*. Springer.
- [4] BERNARDO, J.-M. and SMITH, A. F. M. (2009). *Bayesian theory*. John Wiley & Sons, Ltd., Chichester.
- [5] LINDLEY, D. V. (2014). *Understanding uncertainty*. John Wiley & Sons, Inc., Hoboken, NJ.

[6] MACKEY, D. J. C. (2003). *Information theory, inference and learning algorithms*. Cambridge University Press, New York.

[7] FISHBEIN, D. (2006). *Data analysis*. Oxford University Press, Oxford.

[8] MITCHELL, R. (2016). *Statistical rethinking: A bayesian course with examples in r and stan*. CRC Press.

[9] BOX, G. E. P. and DRAPER, R. (1987). *Empirical model-building and response surfaces*. John Wiley & Sons, Inc., New York.

1.3 The Observational Model

[10] DIACONIS, P. and SKYRMS, B. (2017). *Ten great ideas about chance*. Princeton University Press.

1.4 Model-Based Inferences

[11] BETANCOURT, M. (2015). A unified treatment of predictive model comparison.

2 Frequentist Inference

[12] GOOD, I. (1950). *Probability and the weighing of evidence*. Hafners, New York.

3 Bayesian Inference

[13] TALTS, S., BETANCOURT, M., SIMPSON, D., VEHTARI, A. and GELMAN, A. (2018). Validating bayesian inference algorithms with simulation-based calibration.

4 Comparing Asymptotic Apples to Preasymptotic Oranges

[14] STAN DEVELOPMENT TEAM. (2018). Stan: A C++ library for probability and sampling, version 2.17.1.

5 Conclusion

Acknowledgements

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References

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Original Computing Environment

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## Original Computing Environment

**sessionInfo()**

R version 3.5.1 (2018-07-02)

Platform: x86\_64-apple-darwin15.6.0 (64-bit)

Running under: macOS Sierra 10.12.6

Matrix products: default

BLAS: /Library/Frameworks/R.framework/Versions/3.5/Resources/lib/libRblas.0.dylib

LAPACK: /Library/Frameworks/R.framework/Versions/3.5/Resources/lib/libRlapack.dylib

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attached base packages:

[1] stats graphics grDevices utils datasets methods base

loaded via a namespace (and not attached):

[1] compiler\_3.5.1 backports\_1.1.2 magrittr\_1.5 rprojroot\_1.3-2

[5] tools\_3.5.1 htmltools\_0.3.6 yaml\_2.2.0 Rcpp\_0.12.19

[9] stringi\_1.2.4 rmarkdown\_1.10 knitr\_1.20 stringr\_1.3.1

[13] digest\_0.6.18 evaluate\_0.12